IE802 Notes

Separability

$$* \sum_{i=1}^{n} |x_i| \leq 10$$

$$x_1 + x_2 \cdots x_n \leq 10$$
$$-x_1 + x_2 \cdots \leq 10$$
$$x_1 - x_2 \cdots \leq 10$$
$$\vdots$$
$$-x_1 - x_2 - x_3 \cdots \leq 10$$

2^n constants, $n \rightarrow$ variables

Reformulation

$$\sum_{i=1}^{n} w_i \leq 10, \quad |x_i| \leq w_i \; i = 1, 2, \ldots n$$

$$\Leftrightarrow \sum_{i=1}^{n} w_i \leq 10, \quad x_i \leq w_i \; i = 1, 2, \ldots n, \quad -x_i \leq w_i \; i = 1, 2, \ldots n$$

$2n + 1$ constraints, $2n$ variables

$$\rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 \leq 100$$

Scheme Say point $(10, 0, 0, 0, 0)$: $F(\hat{x}) + \nabla \cdots - 100 \leq 0$

pt $(0, 10, 0, 0, 0)$: $\hat{F}(\hat{x}) + \nabla \cdots - 100 \leq 0 \rightarrow$ Linearization

pt ... pt

[Diagram: pt-related sketch — a circle intersected by lines, labeled "pt sqd" (squared).]

- Both schemes are equivalent → more tighter than above; any point that satisfies above also satisfies this

Scheme

$$w_1 + w_2 + w_3 + w_4 + w_5 \leq 100$$
$$x_1^2 \leq w_1,\ x_2^2 \leq w_2,\ x_3^2 \leq w_3,\ x_4^2 \leq w_4,\ x_5^2 \leq w_5$$

$\widehat{x}$ (10,0,0,0,0); $\widehat{w}$ (100,0,0,0,0)

$$x_1^2 \leq w_1 \to x_1^2 - w_1 \leq 0 \to 0 + [20\ -1] \begin{bmatrix} x_1 - 0 \\ w_1 - 100 \end{bmatrix} \leq 0$$

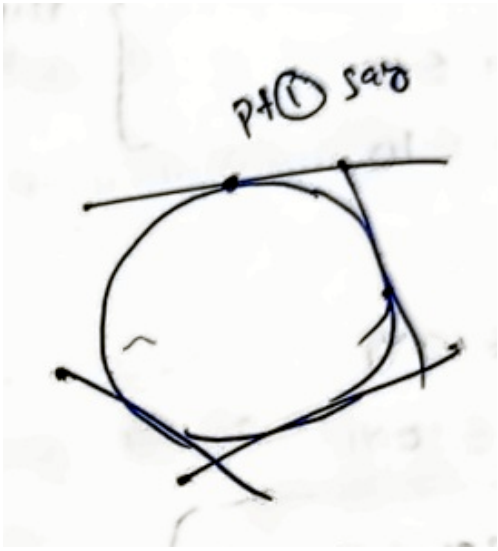


Figure 1: a rectangle with diagonal, labeled $F(\widehat{x}) + \nabla \cdots$

$\Rightarrow f \leq 0$

$$f = f_1(x_1, x_2, x_3) + f_2(x_4, x_5, x_6) + f_3(x_4, x_8, x_9) + f_4(\dots)$$

Linearise f at k points $\to$ weak

$$f_1(x_1, x_2, x_3) \leq w_1 \to \exists \lim,\ w_2 \to 3 \text{ linearizations}$$

$w_1,\ w_4$

equivalent to $w_1 + w_2 + w_3 + w_4 + w_5 \leq 0$ [as] $3^4 = 81$ linearization in original

17/02, 2nd page (50 hrs → 1 sec)

- Linearization makes calculation very fast

$$\min(x_1 + x_2 - x_3 + 4x_4)^2 \to \text{Not separable}$$
$$\min(x_1 - 2x_2 + x_3)^2 + (x_4 + x_5 - x_7)^2 + x_6^2$$

$$** \min f_1(x_1, x_2, x_3) + f_2(\cdots) + f_3(\cdots) + f_4(x_{10}, x_{11}, x_{12})$$

$$\text{st } g_i(x, y) \leq 0$$

$$\min w_1 + w_2 + w_3 + w_4$$
$$f_1 \leq w_1, \quad f_4 \leq w_4, \quad g_i(x, y) \leq 0$$

Non-linear b & b; Linearization not suitable fast, non-linear

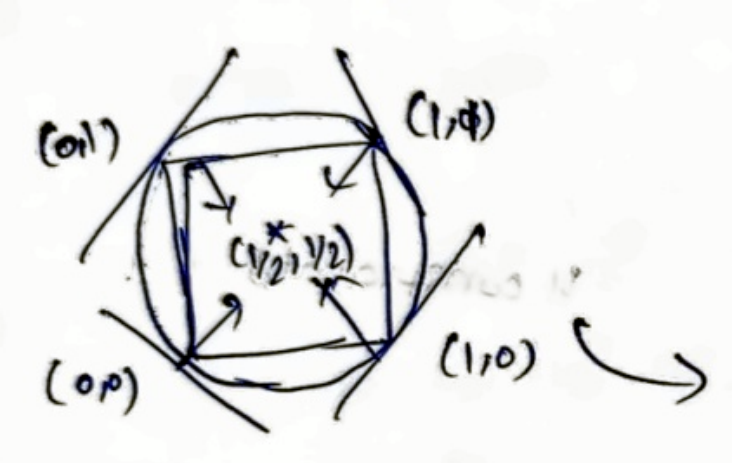


Figure 2: A square inscribed in a circle, corners labeled $(0, 1)$, $(1, 1)$, $(1, 0)$, $(0, 0)$, center $(1/2, 1/2)$ — a sphere in which there are no integer points, made square as relaxation.

Quesada and Grossman are slow in solving this, King separability is fast; complexity → non-separable is separable expl(oit)?

Automatic Reformation (solvers use this)

$$* \min e^{x_1+x_2} + 5e^{x_3} + 7e^{x_4+x_5} - \log(x_2 + x_4)$$

(Only 2 separations we can make) : variable in one separation should not be in another separation.

Perspective Reformation:



Figure 3: a circle containing a smaller shaded triangle/wedge shape.

→ shrinking the region from known feasible points to make it more tighter and faster

19/02/26

Automatic Reformulation.

$$\text{model-M} \downarrow \text{ Reformulate model-}\hat{m}$$

Perspective Reformulation - Motivation from linear models: e.g. model this relation (ex-Factory problem)

$y_1 = 0 \Rightarrow x_1 = 0 \quad y_1 = 1 \Rightarrow x_1 \in [0, 5]$

Soln: $0 \leq x_1,\ x_1 \leq 5y_1 \quad **, \quad y_1 \in \{0, 1\},\ x_1 \in \mathbb{R}$

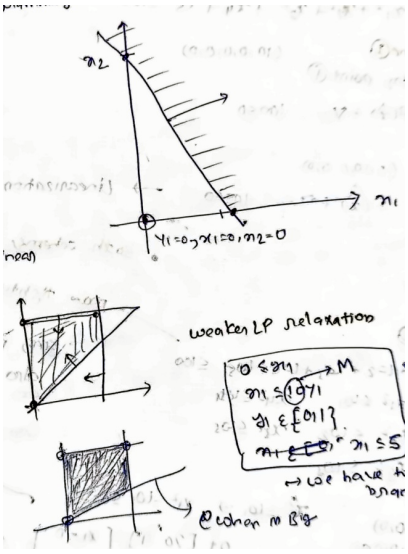


Figure 4: A 2D region (x_1, x_2 axes) — a shaded triangular wedge above a line from origin, with point $y_1 = 0, x_1 = 0, x_2 = 0$ marked at origin, illustrating the disjunctive feasible region.

ex. $y_1 = 0 \Rightarrow x_1 = 0$ and $x_2 = 0$ $\left.\begin{array}{l} y_1 = 1 \Rightarrow 3x_1 + x_2 \geq 10 \\ -10 \leq x_1 \leq 10 \\ -10 \leq x_2 \leq 10 \end{array}\right\}$ Disjunctive Programming

Soln:

$$-10y_1 \leq x_1 \leq 10y_1$$
$$-10y_1 \leq x_2 \leq 10y_1$$
$$\boxed{M(1 - y_1) + 3x_1 + x_2 \geq 10} \leftarrow \text{very Big number} \approx 1000$$
$$\boxed{y_1(3x_1 + x_2) \geq 10y_1} \rightarrow \text{Non-linear}$$

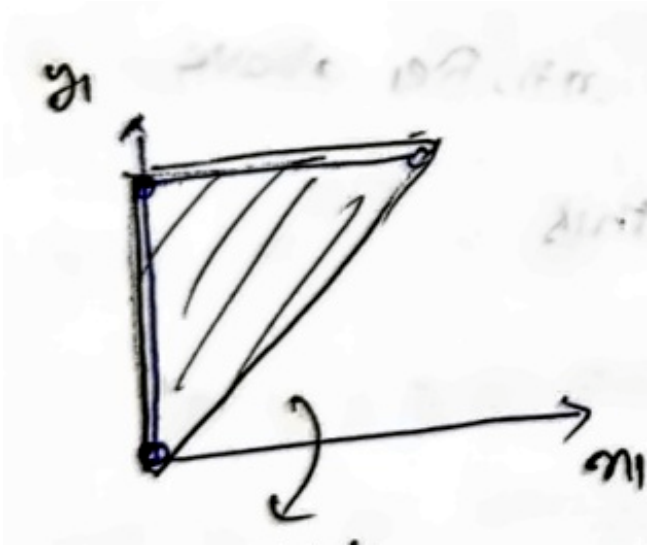


Figure 5: A shaded triangular region in (y_1, x_1) space, "Tight LP relaxation" ***, followed by two more sketches of shaded triangular/quadrilateral relaxation regions, and a box: weaker LP relaxation — $-M \leq x_1$, $x_1 \leq 10y_1$, $y_1 \in \{0, 1\}$, `x_1\in\[0,5\]` (not typeset) → we have to solve branches, "c(hoos)e M big".

model-: $x_1^2 + x_2^2 \leq 100$, $-10y_1 \leq x_1 \leq 10y_1$, $-10y_1 \leq x_2 \leq 10y_1$, $y_1 \in \{0, 1\}$

Ex $y_1 = 1 \Rightarrow \begin{cases} x_1^2 + x_2^2 \leq 100 \\ -10 \leq x_1, x_2 \leq 10 \end{cases}$ $\quad y_1 = 0 \Rightarrow x_1 = x_2 = 0$

$$x_1^2 + x_2^2 \leq 100y_1$$
$$-10y_1 \leq x_1 \leq 10y_1, \quad \text{tighter}$$
$$-10y_1 \leq x_2 \leq 10y_1$$
$$y_1 \in \{0, 1\}$$

Relaxation 2 $\leq$ Relaxation-1

- When $y_1 \in \{0, 1\}$ → Model 1 and model 2 have same feasible region
- When $0 \leq y_1 \leq 1$ then are they same? (Variable)

$y_1 = 1/2 \Rightarrow x_1 = ?, x_2 = ?$

 $x_1^2 + x_2^2 \leq 100 y_1^2$ → strictly tighter → but no more convex ⋮ $y_1 = 1/2$, $x_1 = 5$, $x_2 = 5$ → Not in relaxation

 $\sqrt{x_1^2 + x_2^2} \leq 10 y_1$ ← convex; not allowed b/c $y_1 \in \{0, 1\}$ always $+ve$

→ No derivative @ (0,0) problem

Defn: Let $g(x_1, x_2, x_3 \ldots x_n)$ be a convex function. Then its perspective is defined as the fn:

$$\bar{g}(x_1, x_2 \ldots x_n, t) = \begin{cases} t g\left(\frac{x_1}{t}, \frac{x_2}{t}, \ldots, \frac{x_n}{t}\right) & \text{when } t \neq 0 \\ 0 & \text{when } t = 0 \end{cases}$$

Use: Given constraint $g(x_1, x_2, x_3 \ldots x_n) \leq 0$; tighten it to $y_1 g\left(\frac{x_1}{y_1}, \frac{x_2}{y_1} \ldots \frac{x_n}{y_1}\right) \leq 0$

$$y_1 l_i \leq x_i \leq y_1 u_i \quad l_i \leq x_2 \leq u_2 y_1$$

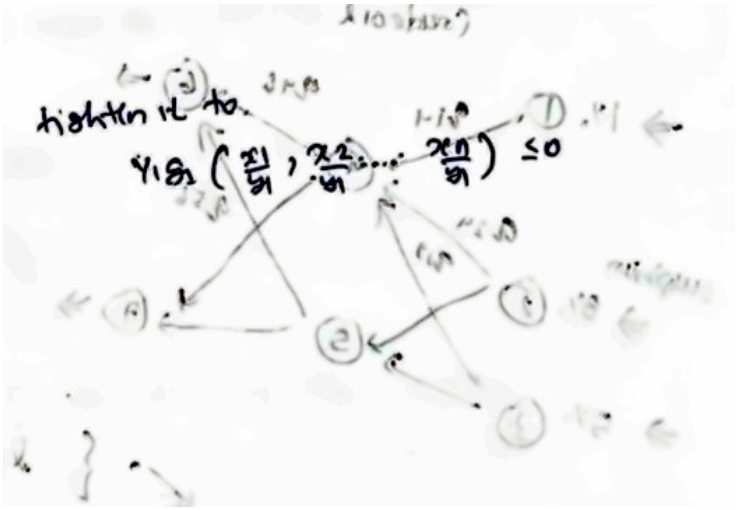


Figure 6: A pentagon-like directed graph with 5 nodes labeled -, connected by arrows, illustrating some kind of network/ordering relation, faint and undecipherable in detail.

model-1

$$x_1^2 + x_2^2 \leq 100$$
$$x_1^2 + x_2^2 - 100 \leq 0$$
$$-10 x_1 \leq x_1 \leq 10 x_2$$

Try! 1. Taking function $g_1(x_1, x_2) = x_1^2 + x_2^2 - 100$ 2. Apply perspective formulae: $y_1 \left[\left(\frac{x_1}{y_1}\right)^2 + \left(\frac{x_2}{y_2}\right)^2 - 100\right] \leq 0$ 3. Distribute sq. $y_1 \left[\frac{x_1^2}{y_1^2} + \frac{x_2^2}{y_2^2} - 100\right] \leq 0$ 4. Multiply the outer y_1 through the terms: $\frac{x_1^2}{y_1} + \frac{x_2^2}{y_1} - 100 y_1 \leq 0$

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For convex MINLO we don't know the function

$$\min f(x,y)$$
$$g_i(x,y) \le 0,\ i = 1,2 \dots m$$
$$x \in \mathbb{R}^n, \quad y \in \mathbb{Z}^p$$

$$g_i(\hat{x},\hat{y}) + \nabla g_i(\hat{x},\hat{y}) \begin{bmatrix} x-\hat{x} \\ y-\hat{y} \end{bmatrix} \le 0 \quad \rightarrow \text{solve by} \begin{cases} (1)\ \text{Branch and Bound} \\ (2)\ \text{Convex Relaxation} \\ (3)\ \text{Interval analysis} \end{cases}$$

Global Optimization of "Factorable Functions"

Recursive fn using a limited no. of operators: $+, -, \times, \div, \log,\ ^\wedge k,\ k^{()}, \sin, \cos, \tan(c) \dots$

Ex: $x_1^{()} * (3x_1^2 + \log(5 + \sin^2(5x_1x_2 + 7)))$

Ex- Powerflow problem **Pooling applications**

Crudeoil: [Diagram: A network flow diagram with source nodes □(1%), □(8%, "sulphon"), □(5%) feeding into intermediate nodes □, □, which feed into output nodes □, □, with edge flows labeled $q_{14}, q_{24}, q_{34}, q_{46}, q_{56}$ etc.]

$$p_4 = \frac{0.01q_{14} + 0.08q_{24} + 0.05q_{34}}{q_{14} + q_{24} + q_{34}}, \qquad p_6 = \frac{q_{46} \times p_4 + q_{56} \times p_5}{q_{46} + q_{56}}$$

$\{26 < p_6 < 46\}$ — constraint on supply, flow, etc. There is upper and lower bound; more sulphur → more benefit

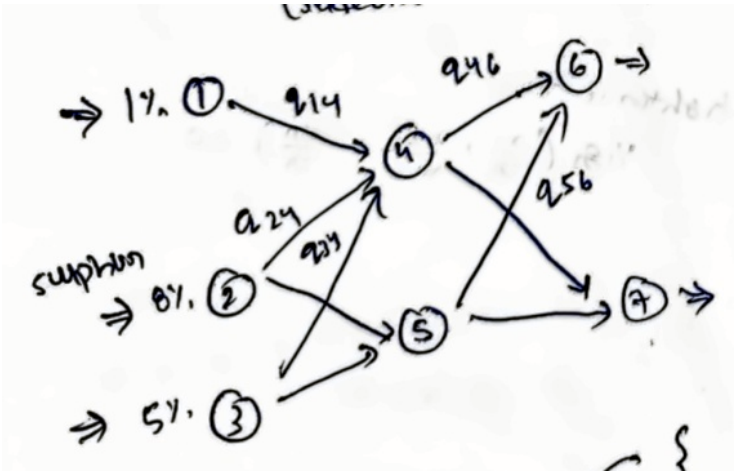


Figure 7: drawing from the page (no description in the notes)

Methods for Global Optimization:

Ex: $\min x_1 - 2x_2$

$$\text{st } x_1^2 + 2x_2 \le 0$$

$$-x_1^2 + 4x_2 \leq 4$$
$$x_1, x_2 \in \mathbb{R}$$

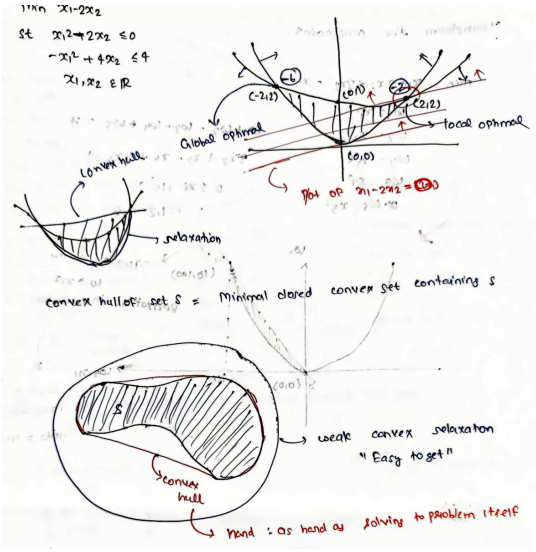


Figure 8: A curve with local minimum marked at $(-2, 2)$ labeled "Global optimal", a saddle-like point at $(0, 1)$, and another critical point at $(2, 2)$ labeled "local optimal", with a red line through $(0, 0)$ representing the level set "λ of $x_1 - 2x_2 = (0)$".

$\rightarrow$ **Convex hull**: [Diagram: a bowl-shaped curve with a shaded region beneath a chord, labeled "relaxation".]

ex: Convex hull of set S = minimal closed convex set containing S

ex: [Diagram: An irregular blob shape S shaded, enclosed within its convex hull (a larger smooth oval outline), with the gap between them labeled "convex hull".]

→ weak convex relaxation "Easy to get"; hard: as hard as solving to problem itself

How to get these weak convex relaxations?

Ex: Min $x_1 + 2x_2 + 3x_3 - x_4$

$$\text{st } x_1^2 + x_2^2 - x_3^2 + x_5^2 - x_4^2 = 7 \Big\} \text{Non-linear + Non-convex}$$

$$3x_2 + x_3 - x_4 + 3x_5 = 1$$
$$0 \leq x_i \leq 10, \quad i = 1, 2 \ldots 5$$

Transform the constraint.

min $x_1 + 2x_2 + 3x_3 - x_4$

$w_1 = x_1^2$, $w_2 = x_2^2$, $w_3 = x_3^2$, $w_4 = x_4^2$, $w_5 = x_5^2$

$$w_1 + w_2 - w_3 - w_4 + w_5 = 7$$
$$3x_2 + x_3 - x_4 + 3x_5 = 1$$

$$0 \leq x_i \leq 10,\ i = 1, 2 \dots 5$$

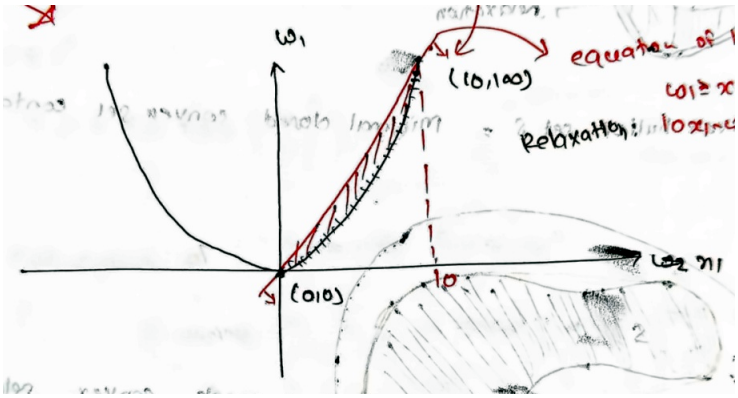


Figure 9: A bowl-shaped curve w_1 vs x_1 (parabola $w_1 = x_1^2$), with the region between the curve and its chord from $(0, 0)$ to $(10, 100)$ shaded — "relaxation" — labeled "equation of line".

Relaxation: $x_1^2 \leq w_1$, $0 \leq x_1 \leq 10$, i $= 1, 2 \dots 5$; $w_2 \geq x_2^2$, $0 \leq x_2 \leq 10$, adds 5 times.

[Diagram: A large irregular blob shape labeled S with a smooth convex-hull outline around it, both shaded.]

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$$\min x_1 + 2x_2 + 3x_3 - x_4$$

□ st $x_1^2 + x_2^2 - x_3^2 - x_4^2 + x_5^2 = 7 \rightarrow$ Non convex

$$3x_2 + x_3 - x_4 + 3x_5 = 1$$

$$0 \leq x_i \leq 10,\ i = 1 \dots 5$$

Transform: □

$w_1 + w_2 - w_3 - w_4 + w_5 = 7$ $w_1 = x_1^2$ $w_2 = x_2^2$ $w_3 = x_3^2$ $w_4 = x_4^2$ $w_5 = x_5^2$ $0 \leq x_i \leq 10,\ i = 1 \dots 5$

To put a lid, we need to know the bounds of x_i; we want to be online

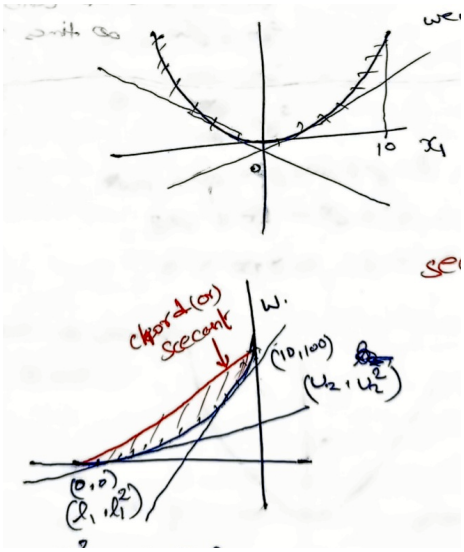


Figure 10: Parabola $w = x_1^2$ over `\[0,10\]` (not typeset), with a red chord/secant labeled "chord (or secant)" from the origin to $(10, 100)$, and a second sketch showing a tangent line at (u_2, u_2^2) touching the curve near $(0, 0)$ and (l_1, l_1^2) — "secant works only for square types".

equality, $x_1^2 - w_1 = 0$

$$\boxed{g(\hat{x}) + \nabla g(\hat{x})^T (x - \hat{x}) \leq 0} \leftarrow \text{This doesn't depend on bound}$$

at $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow 0 + [0 \; -1] \begin{bmatrix} x_1 - 0 \\ w_1 - 0 \end{bmatrix} \leq 0$

$$w \geq 0$$

for upper bound we can: $\boxed{w_1 \leq 10 x_1}$ — This depends on bounds

Secant was not needed in Convex MINLP

We solve (P) □. We relax the programme:

LP relaxation

$$\text{Min } x_1 + 2x_2 + 3x_3 - \cdots$$
$$3x_2 + x_3 - x_4 \cdots$$

Secant $\{w_1 \leq 10 x_1$ Tangent $\{w_1 \geq 0, \; w_1 \geq \cdots, \; w_1 \geq \cdots$

$w_2 \geq 0 \; w_2 \leq 10 x_2$

Solve LP relaxation: - if infeasible $\Rightarrow$ (P) infeasible - if the LP optimal point $(\hat{x}, \hat{w})$ satisfies all constraints of (P), then $(\hat{x}, \hat{w})$ is optimal to (P). - LP relaxation is unbounded $\rightarrow$ [Complaint to the modeller] $-1 \leq x_1 \leq 1$, $\frac{1}{x_1}$ - $(\hat{x}, \hat{w})$ does not satisfy all non-linear constraints of (P), we have 2 cases in this style.



Figure 11: A crescent/wedge-shaped feasible region with point $(\hat{x}, \hat{w})$ marked inside — "we can get such a point in simplex".

 There are lot of variables in high dimension, such cases happen via simplex.

But not on the curve (so) 2 things happen: 1) Above the curve: 2) Below the curve: if $\hat{w}_1 < \hat{x}_1^2$, add a tangent at $(\hat{x}_1, \hat{x}_1^2)$

We will find a point on curve, to set tangent 1) by drawing a line vertically. 2) try to find nearest point on curve to get the tangent.

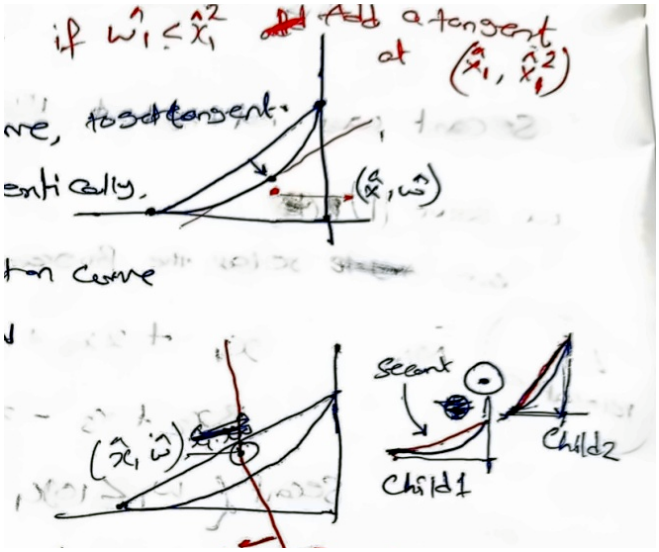


Figure 12: Two curve sketches showing a point $(\hat{x}, \hat{w})$ inside the region between the curve and a chord, with a red vertical line and a labeled tangent construction; a second small diagram shows "secant", "Child1", "Child2" nodes.

1) **Above the curve:**

We do branch & bound. Child | Child

→ We will draw a secant that cut points in both splitted individual childs.

All we need is, we don't count that point & we don't want to lose our original feasible region (curve).

→ at the corners, the secant cuts off the point because the secant is tight at corners.

Here there is no integer requirements, but we do branch & bound, there will be many cases (or branches) as not integer.

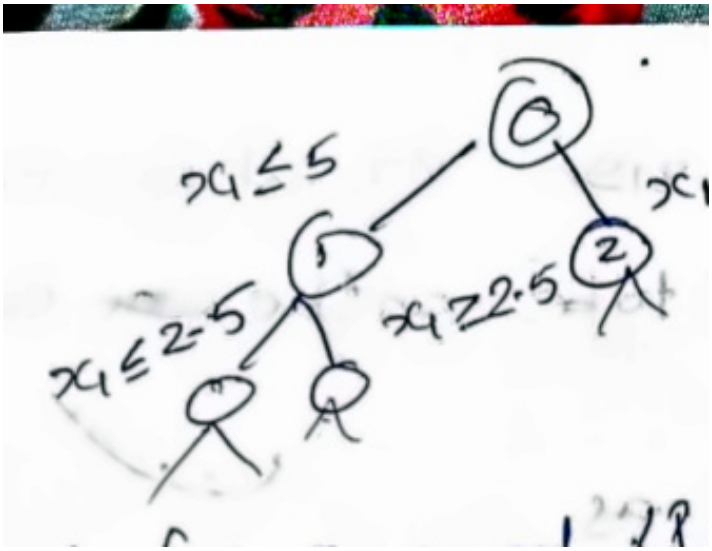


Figure 13: A binary branching tree — root node $x_1 \leq 5 \mid x_1 \geq 5$ splitting into two children, each further split at $x_1 \leq 2.5$ / $x_1 \geq 2.5$ — "Same branch & bound techniques & LP rules".

That's all for 5 round LPs. Add variables if difficulty.

$$\min e^{-x_1^2+x_2}$$
$$\text{st } x_1 + 2x_2 = 1$$
$$-10 \leq x \leq 10$$

Transform ⇒ (non-convex constraint) $w_1 = x_1^2$, $w_2 = x_2 - w_1$, $w_3 = e^{w_2}$

due to minimization both will become same, $w_3 \geq e^{w_2}$ — convex constraint

$e^{-x_1^2+x_2}$ is not completely convex, but looks like normal dist.

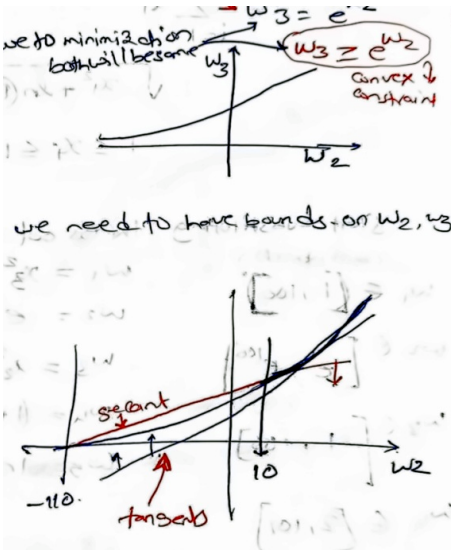


Figure 14: An exponential curve $w_3 = e^{w_2}$ rising from flat near 0 to steep, over axis w_2.

We don't have bounds, we need to have bounds on w_2, w_3

$$x_1 \in [-10, 10], \quad w_1 \in [0, 100], \quad w_2 \in [-110, 10], \quad w_3 \in [e^{-110}, e^{10}]$$

We need to add variables to make linear, so we set find bounds & easy.

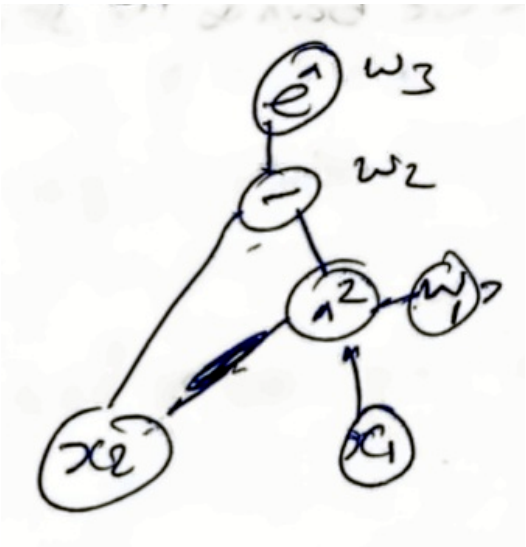


Figure 15: A small tree: $e^{w_3} \leftarrow -w_2 \leftarrow x_2 - x_1^{()2}$, with leaves x_1, x_2.

There is relation b/w Computational graph & Constraints.

We store nonlinear equations in computational graphs, for linear equations we use linear functions.

$$w_3 = e^{w_2}$$

due to $=$, it is not convex, as we have to be on the curve.

```
w_3\geq e^{w_2} \quad \text{this is convex, as we can have feasible region on & above
↪  curve}
```

Warning: equation could not be typeset; shown as written.

Ex:

$$\min \sum_{i=1}^{5} x_i$$

$$\sqrt{x_1^2 + \ln(1 + x_2^2)} \leq e^{\frac{x_2^2}{x_3}} + 4x_4 + x_5$$

$$1 \leq x_i \leq 10 \ \forall i$$

Start substituting things out. (closed intervals)

$w_1 \in [1, 100]$ $w_2 \in [e, e^{100}]$ $w_3 \in [1, 100]$ $w_4 \in [2, 101]$ $w_5 \in [\ln 2, \ln(101)]$ $w_6 \in [1, 100]$

$$w_1 = x_1^2, \quad w_2 = e^{w_1}, \quad w_3 = x_2^2, \quad w_4 = (1 + w_3), \quad w_5 = \ln(w_4)$$
$$w_6 = x_1^2, \quad w_7 = w_6 + w_5, \quad w_8 = \sqrt{w_7}, \quad w_8 \leq w_2 + 4x_4 + x_5$$

← we use bounds for secants



Figure 16: drawing from the page (no description in the notes)

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$$w_1 = x_1^2, \ l_1 \leq x_1 \leq u_1$$

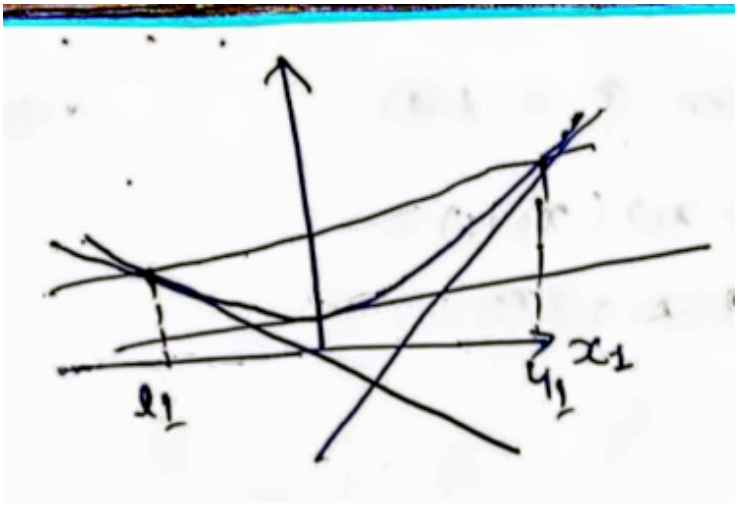


Figure 17: A parabola with a chord between l_1 and u_1 on the x-axis, showing the secant/tangent construction.

$$w_1 = \log x_1, \quad w_1 = e^{x_1}, \quad w_1 = x_1^3$$

$$\boxed{w_1 = x_1 \cdot x_2}$$

$$\log w_1 = \log x_1 + \log x_2$$

$$w_2 = \log w_1, \quad w_4 = \log x_2, \quad w_3 = \log x_1$$

$$w_1 = x_1 \cdot x_2, \quad 0 \le x_1 \le 1, \quad 0 \le x_2 \le 1, \quad 0 \le w_1$$

$w_1 \le 1 \rightarrow$ drop as convex; $w_1 \le x_1$; $w_1 \le x_2$ — Bilinear constraint

x_1	x_2	w_1
0	0	0
0	1/2	0
1/2	1/2	1/4
1	0.8	0.8

$Z = x \cdot y$, $\{x \ge 0, y \ge 0\} \rightarrow$ neither convex nor concave

x_1	x_2	w_1
0	0	0
10	0	0
0	10	0

equation: From image, calculate plane that passes through this point $(0, 1, 0)$, $(1, 0, 0)$, $(1, 1, 1)$

$$w_1 \ge x_1 + x_2 - 1, \quad w_1 \le x_1, \quad w_1 \le x_2, \quad 0 \le w_1$$

New system: $1/2 \le x \le 4$: $(1/2, 1/2, 1/4)$, $(4, 1/2, 2)$, $(1/2, 4, 2)$

$$w_1 \ge 2x_1 + 2x_2 - 4$$
$$0 \le (2 - x_1)(2 - x_2)$$
$$4 - 2x_1 - 2x_2 + x_1x_2 \ge 0$$
$$4 - 2x_1 - 2x_2 + w_1 \ge 0 \to \text{linear constraint}$$

$Z = x * y \to$ trying to cover it: new: $-1 \le x_1 \le 2$, $-1 \le x_2 \le 2$, $-2 \le w_1 \le 4$; □$x_1 \le 2$, □$-1 \le x_1$, □$x_2 \le 2$, □$-1 \le x_2$; □×□ $0 \le 2 - x_1$, □ $0 \le 2 - x_2$. Now multiply co(nstrain)ts.

From □ and □: $(x_1 + 1)(x_2 + 1) \ge 0$, $w_1 + x_1 + x_2 \ge 0$

From □ and □: $(2 - x_1)(x_2 + 1) \ge 0$, $2 - x_1 + 2x_2 - w_1 \ge 0$

also solve □ and □: [similar cross-product]

$$w_1 \ge 2x_1 + 2x_2 + 4$$
$$w_1 \ge -x_1 - 2x_2 - 1$$
$$w_1 \le -x_1 + 2x_2 + 2$$
$$w_1 \le 2x_1 - x_2 + 2$$

$\{x \ge 0, y \ge 0\}$, $Z = xy$

$$3x_1^2 + 4x_2^2 - 7x_1x_2 + 8x_3x_4 + 8x_3 + x_4 \le 10$$
$$x_1^2 = w_1, \quad x_2^2 = w_2, \quad x_1x_2 = w_3, \quad x_3x_4 = w_4$$
$$3w_1 + 4w_2 - 7w_3 + 8w_4 + 8x_3 + x_4 \le 10$$

Let $(\hat{x}, \hat{w})$ be the LP solution with $\hat{w}_3 \ne \hat{x}_1\hat{x}_2$?? → use cut on branch. Not possible bc cutting other points.

$\hat{x}_1 = 1$, $\hat{x}_2 = 1$, $\hat{w}_3 = 1/2 \to$ branch on any one.

Lemma: Let $(\hat{x}_1, \hat{x}_2, \hat{w})$ be any point inside the LP relaxation of $w = x_1x_2$ with $l_1 \le x_1 \le u_1$ and $l_2 \le x_2 \le u_2$.

If $\hat{x}_1 = l_1$ then $\hat{w} = \hat{x}_1\hat{x}_2$. Similarly, if $\hat{x}_1 = u_1$ then... Sim. if $\hat{x}_2 = l_2$ ... If $\hat{x}_2 = u_2$ ...

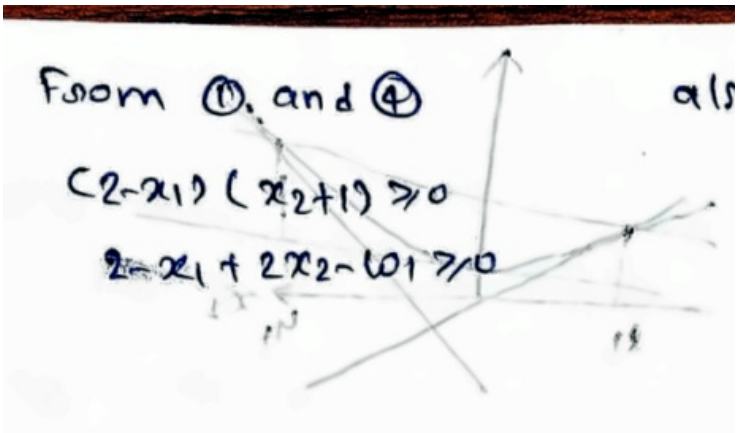


Figure 18: drawing from the page (no description in the notes)

Example: $\dfrac{x_1^2 + x_2^2}{3x_1 + x_3} + 5x_1x_2 \leq 6$

Linearise.

$$x_1^2 = w_1, \quad x_1x_2 = w_2, \quad 3x_1 + x_3 = w_3, \quad \frac{1}{w_3} = w_4, \quad x_2^2 = w_5$$

$$w_6 = w_4 \cdot w_5$$

$$w_1 + w_6 + 5w_2 \leq 6$$

Solve and find $(\hat{x}, \hat{w})$. Check if all satisfies. If anyone not → do branching.

- To draw box around terms, we need bound on x_1, i.e. $\leq x_1 \leq$, $\leq x_2 \leq$ also on $\leq w_1 \leq$, $\leq w_2 \leq \ldots$

Example: Min $x_1 - 10x_2$

$$x_1^2 \leq 2x_2$$

$$-x_1^2 + 4x_2 \leq 4$$

$$-100 \leq x_1, x_2 \leq 100$$

Introduce new variables. LP relaxation

Soln: $w_1 = x_1^2$

$$w_1 \leq 2x_2$$

$$-w_1 + 4x_2 \leq 4$$

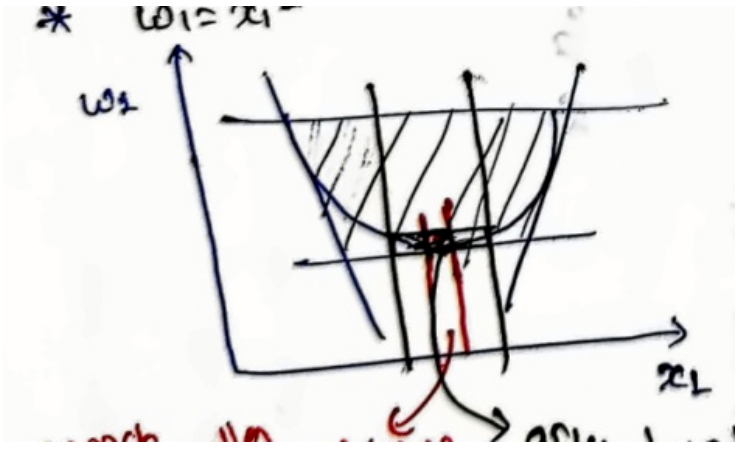


Figure 19: A parabola $w_1 = x_1^2$ with a shaded band around the vertex, labeled "search space smaller" after bound tightening, over range $-2 \leq x_1 \leq 2$.

$$0 \leq w_1 \leq 10{,}000$$

$$-100 \leq x_1 \leq 100$$

Tightening bound from one constraint: $x_1^2 \leq 200$, $x_1 \leq \sqrt{200}$, $-14.1 \leq x_1 \leq 14.1$

After bound tightening, our search space reduces and also secant reduces.

Second const: $-x_1^2 + 4x_2 \leq 4$

$$x_1^2 \geq 4x_2 - 4 \geq -400 - 4 = -404 \rightarrow \text{Not much in B ...}$$

2 constraints at a time:

$$x_1^2 - 2x_2 \leq 0$$
$$-x_1^2 + 4x_2 \leq 4$$
$$\overline{2x_2 \leq 4}$$

gives, $x_2 \leq 2$

Using this in constraint 1: $x_1^2 \leq 2x_2$, $x_1^2 \leq 4$, which gives $\boxed{-2 \leq x_1 \leq 2}$

Ex: $x_1^2 + 5x_1x_2 + 12x_2^2 + 7x_3^2 - 5x_1x_3 \leq 20$, $-1 \leq x \leq 5$

$x_1^2 = w_1$, $x_1x_2 = w_2$, $w_1 \in [0, 25]$ → Not easier to find bound on x_1

Bound on $x \Rightarrow$ Bound on w

Forward bound propagation → tighter bound on

$$x_2^2 = w_3, \quad x_1x_3 = w_4, \quad x_3^2 = w_5$$

$$\boxed{w_1 + 5w_2 + 12w_3 - 5w_4 + 7w_5 \leq 20}$$

also back propagation

$$w_2 \in [-5, 25], \quad w_3 \in [0, 25], \quad w_4 \in [-5, 25], \quad w_5 \in [0, 25]$$

Take col at one side and other, another side → get maximum value of w_1

$$w_1 \leq 20 - w_1 - 5w_2 - 12w_3 + 5w_4 - 7w_5$$

Forward → Reverse, do this multiple times. - adding constraint → problematic for model

Optimization based Bound tightening:

$$\min w_1$$
$$\text{st Linear constraint (outer approx)} + \text{(original linear constraints)}$$

→ one gets lower bound and another gives higher bound..

$$\max w_1$$

Ex: [Diagram: two rectangles labeled “x500 piece” and “x100 piece”, question “How many?”]

- look at single constraint and also by combining…

Ex: $x_1^2 + 2x_1x_2 + x_2^2 \leq 100$

Individual terms → problem, but if we combine: $(x_1 + x_2)^2 \rightarrow$ much tighter bound.

- How to relax ← Global opt...
- How to branch, we have studied...?

What happen when we have relaxation of original problem vs their constraints.

$$\min c^T x$$
$$g_1(x) \leq 0$$
$$g_2(x) \leq 0$$

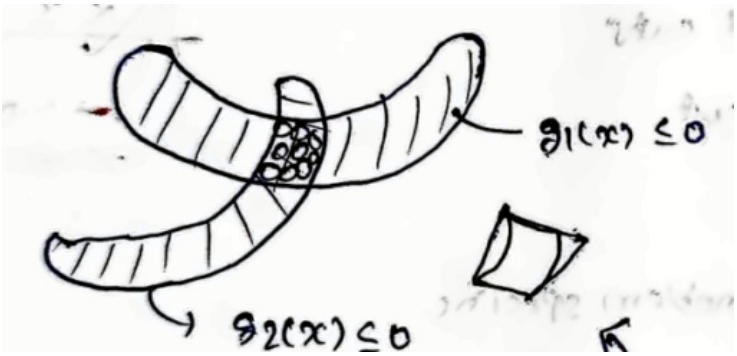


Figure 20: Two irregular boomerang-shaped regions labeled $g_1(x) \leq 0$ and $g_2(x) \leq 0$, overlapping in a small shaded intersection region, and a separate small quadrilateral shape.

Create outer approximation of both.

Ideal: exact relaxation of intersection of two feasible sets.

Normal: Relax $g_1(x)$ and then relax $g_2(x)$ → Take intersection of both relaxations.

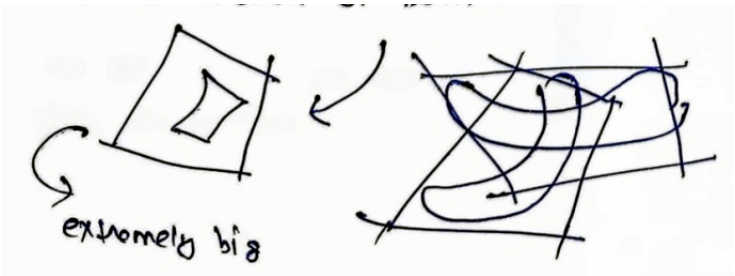


Figure 21: A quadrilateral with a diamond cut out, and a tangled curve shape, labeled "extremely big".

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Relaxation of intersection of 2 non-convex sets tighter than Intersection of relaxations.

Let S_1, S_2 be two non convex sets, then take intersection.

$$\text{Convex Hull}(S_1) \cap \text{Convex hull}(S_2) \supseteq \text{Convex hull}(S_1 \cap S_2)$$

Convex hull: in any non-convex set, the minimal convex set that can fit in [contain the] non-convex set.

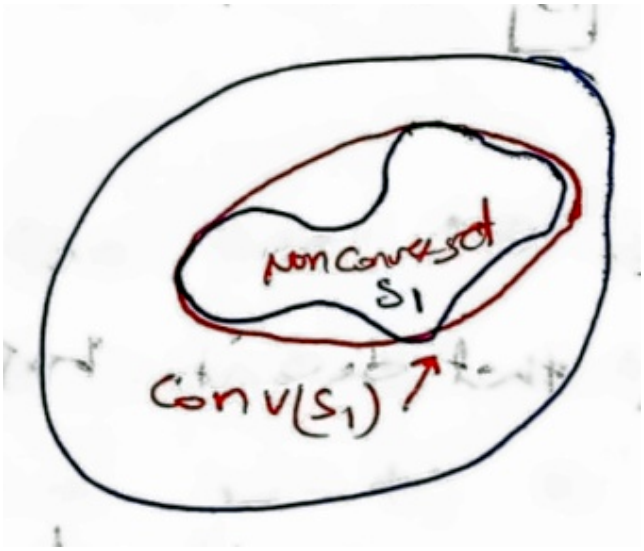


Figure 22: A large oval labeled "Conv(S_1)" (blue outline) enclosing an irregular blob labeled "non convex set S_1" (red outline).

Proof: Pick any point in RHS. It is in LHS (Conv(S_1) & Conv(S_2))

we get 3 constraints. $g_1(x) \leq 0 \leftarrow$ feasible region of 1st constraint $= S_1$ $g_2(x) \leq 0 \leftarrow$ 2nd $\cdots = S_2$ $g_3(x) \leq 0 \leftarrow$ 3rd $\cdots = S_2$

Ideally: find Conv($S_1 \cap S_2 \cap S_3$) → It is hard & complicated, this is much more difficult than finding original answer.

Separately find convex hull of S_i individually for all, then intersect.

Reality: $\text{Conv}(S_1) \cap \text{Conv}(S_2) \cap \text{Conv}(S_3)$

Can we make good tight relaxation by looking at all constraints rather than doing term by term relaxation.

$$x_1^2 - x_2^2 + 3x_1x_2 \leq 5$$
$$x_1^2 = w_1, \quad x_2^2 = w_2, \quad x_1x_2 = w_3$$
$$w_1 - w_2 + 3w_3 \leq 5$$

We know how to relax individual constraints but how to do on whole.

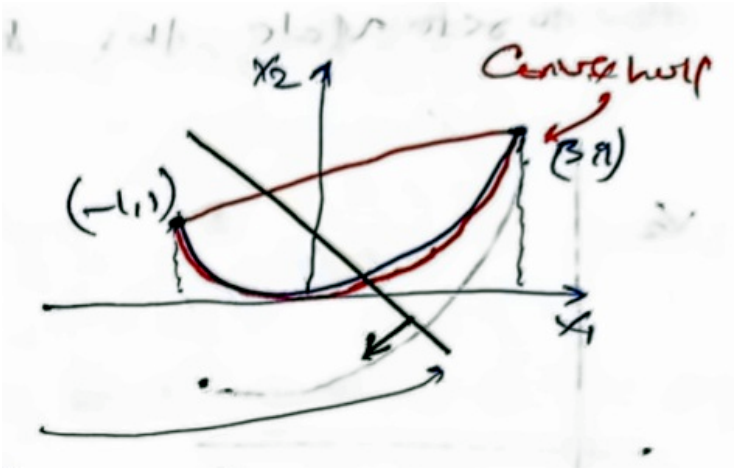


Figure 23: A parabola-like curve in (x_1, x_2) plane from $(-1, 1)$ to $(3, 9)$, with a red "Convex hull" line connecting the endpoints, showing the true curve tightly bounded by the hull.

Ex 1: $\square\ x_2 = x_1^2, -1 \leq x_1 \leq 3\ \}\ S_1$

if I take new constraint $x_1 + x_2 \leq 1\ S_2$

$S_1 \cap S_2 \rightarrow$ [shaded lens shape] not tight as desired is [half-moon shape]

→ we do bound tightening. $x_2 \in [0, 9]$

from $(x_1 + x_2 \leq 1)$ the x_1 can have bound: Upper bound of 1. Now we update to $\boxed{-1 \leq x_1 \leq 1}$ this gives better result.

$\square\ x_1 x_2 \geq 40$

$$0 \leq x_1 \leq 10$$
$$0 \leq x_2 \leq 10$$

add a new variable. $w_1 = x_1 x_2 \rightarrow w_1 \geq 40$, $0 \leq x_1 \leq 10$, $0 \leq x_2 \leq 10$ — outer box using bilinear relaxation

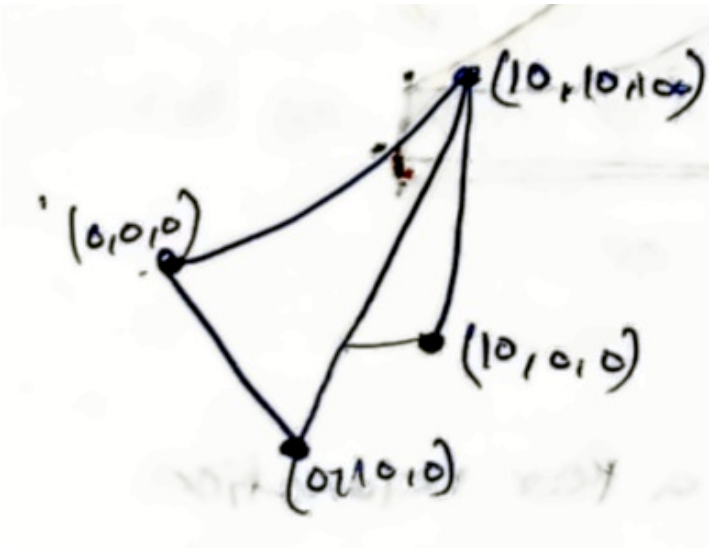


Figure 24: A 3D plot with vertices $(0, 0, 0)$, $(0, 10, 0)$, $(10, 0, 0)$, $(10, 10, 100)$ forming a tetrahedron-like shape.

we get a curve where cut at 40, still non convex in 1 direction & reverse other side.



Figure 25: two boomerang/cone shaped sketches.

Things get complicated, just adding a new bound: "Conv($S_1 \cap S_2$)" — doable but not simple.

$x_1 \cdot x_2$ is not convex expression, but when we draw it, it is convex.

How to reformulate this so the expression will also be convex.

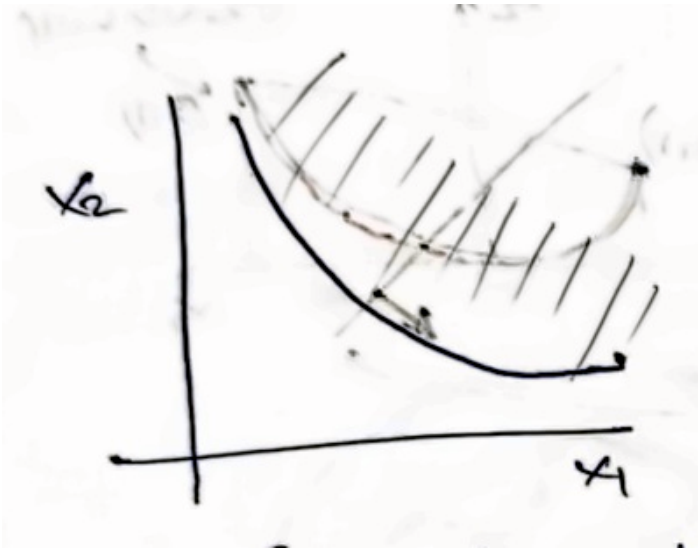


Figure 26: A decreasing hyperbola-like curve in (x_1, x_2) with shaded region above it, labeled "we can validate $x \geq 1, y \geq 1$".

lot of the convex set, but just by looking at constraint $x_1 \cdot x_2$ which is not convex, we cannot say everything is non convex.

Reformulation:

$$x_1 \geq \frac{40}{x_2} \quad \text{(or)} \quad \frac{40}{x_2} - x_1 \leq 0 \leftarrow \text{convex function}$$

All we have to do it by taking tangents all way curve.

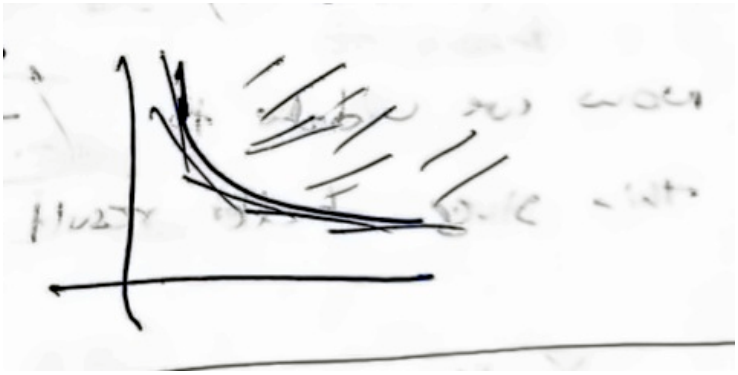


Figure 27: A decreasing hyperbola with several tangent lines drawn along it.

$$x_1 x_2 \leq 40, \quad 0 \leq x_1 \leq 10, \quad 0 \leq x_2 \leq 10$$

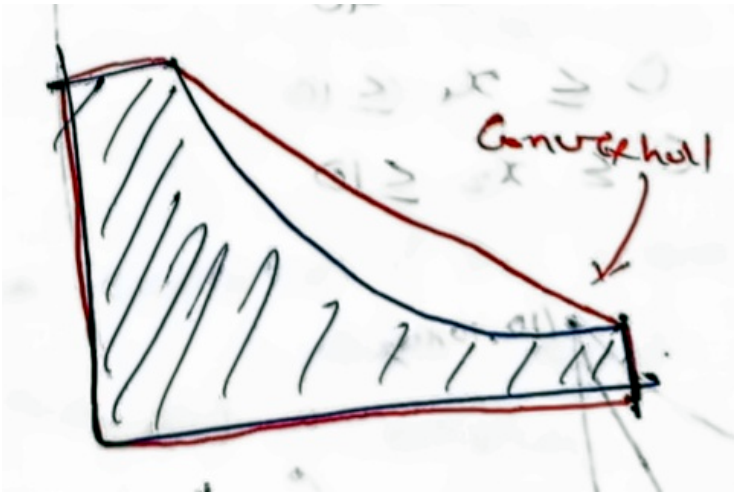


Figure 28: A shaded quadrilateral wedge region with a red curve labeled “Convex hull” along its upper-right edge.

If we do $w_1 \leq 40$, $w_1 = x_1 x_2 \leftarrow$ this will give a floor relaxation. Not always things get simpler.

$\square\ 3x_2 - 2x_3 + x_4 - 4x_5 \geq \dfrac{y_1}{1 + x_1 + x_{10}}$

$$50 \geq x \geq 0$$
$$y \in \{0, 1\}$$

y_1 is binary.

What will be good relaxation.

$$\frac{(3x_2 - 2x_3 + x_4 - 4x_5)}{w_1} \underbrace{(1 + x_1 + x_{10})}_{w_2} \geq y_1$$

$$w_1 \cdot w_2 \geq y_1$$

we can come up with relaxations to different cases of $y = 0, y = 1$.

Trick: $y_1 \in \{0, 1\} \Leftrightarrow y_1 = y_1^2$

add a square to y: $\rightarrow y_1^2$

can try adding y_1^2, this makes constraint convex. We can induce convex.

$$H/W: \quad w_1 \geq \frac{x_1^2}{x_2}$$

Trying to see some things works.

$$\text{Min } y_1^2 + y_2^2 - y_3^2 + 7y_4^2 - 8y_5^2$$

$$Ay = b$$
$$y \in \{0,1\}^p$$

Add squares to make nonlinear to linear integer model

$$y_1 + y_2 - y_3 + 7y_4 - 8y_5$$

If we have more variable it will be Non Convex with/without squares.

$$y_1^2 + y_2^2 \pm y_3^2 \cdots + y_1y_2 - 3y_2y_3 + y_4y_5 \quad \text{not positive definite}$$

$\rightarrow$ Min $\frac{1}{2}y^TQy + c^Ty \leftarrow$ equivalent concave model

$$Ay = b$$
$$y \in \{0,1\}^p$$

We can always add & subtract $y_i^2 - y_i$, both are same so we get original model.

$$\min \frac{1}{2}y^TQy + c^Ty + \sigma_1y_1^2 + \sigma_2y_2^2 + \ldots \sigma_ny_n^2 - \sigma_1y_1 - \sigma_2y_2 - \cdots - \sigma_ny_n$$

Convex? if choose σ to be big enough, this expression will be convex.

$$\boxed{\sigma \geq |\text{most negative (or) smallest eigen value of } Q|} \rightarrow \text{mod}$$

Solve QP then branch, then QP then branch.

$$\min y_1^2 + y_2^2 - y_3^2 + 7y_4^2 \rightarrow +y_3^2 - y_3 \text{ now convex.}$$

1) $\underbrace{x_1x_2}_{w_1} + \underbrace{x_2x_4}_{w_2} + \underbrace{x_1x_3}_{w_3} + x_1x_4 + x_1x_5 \leq 100$

2) $\dfrac{x_1x_4}{w_1} + x_1\underbrace{(x_2 + x_3 + x_4 + x_5)}_{w_3} \leq 100$

option 2) is best as we have 2 bilinear terms.

Consider $0 \leq x_i \leq 10$

$w_3 = x_2 + x_3 + x_4 + x_5$, $w_4 = w_3x_1$

$$0 \leq x_1 \leq 10$$

$0 \leq w_3 \leq 40 \leftarrow$ The bounds are bigger, no guarantee it is tight or not — can be bigger set (or) weak

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$$\min f(x,y)$$

$$\text{st } g_i(x,y) \leq 0$$
$$x \in \mathbb{R}^n,\ y \in \mathbb{Z}^p$$

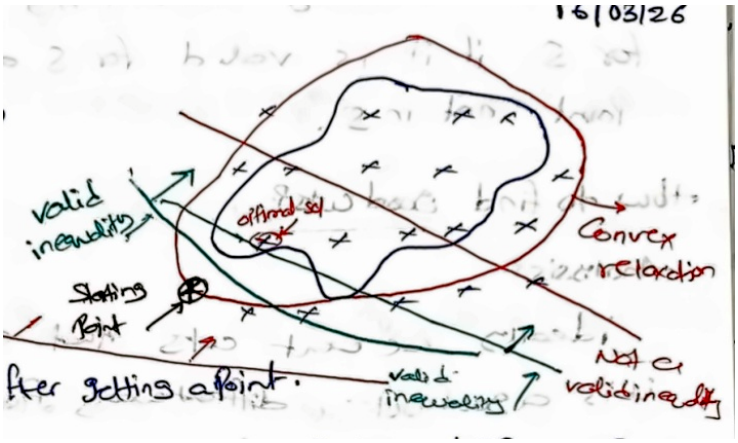


Figure 29: A red blob region overlapping with a blue irregular region full of scattered “x” points, with “valid inequality” lines drawn tangent to the boundary, one line labeled “optimal sol” and another marked “not a valid inequality”; region labeled “Convex relaxation”.

Branch (or) [not a useful cut] make relaxation tighter after getting a point. Starting point.

By branching the work increases as 1 problem becomes 2 & more common.

→ The better option is to tighten relaxation after we have an initial solution of relaxation.

→ before initial point we can’t say is it tight or not.

→ The new relaxation is convex even by having a cut which is valid inequality.

→ A cut which not cuts feasible points is also valid inequality.

→ if valid inequality cuts off optimal point then it is not a valid inequality.

Given a model

(P) $g_i(x,y) \leq 0, i = 1, \dots, m$

$$\min f(x,y)$$
$$x \in \mathbb{R}^n,\ y \in \mathbb{Z}^n$$

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$$\min f(x,y)$$
$$\text{st } g_i(x,y) \leq 0$$
$$x \in \mathbb{R}^n,\ y \in \mathbb{Z}^p$$

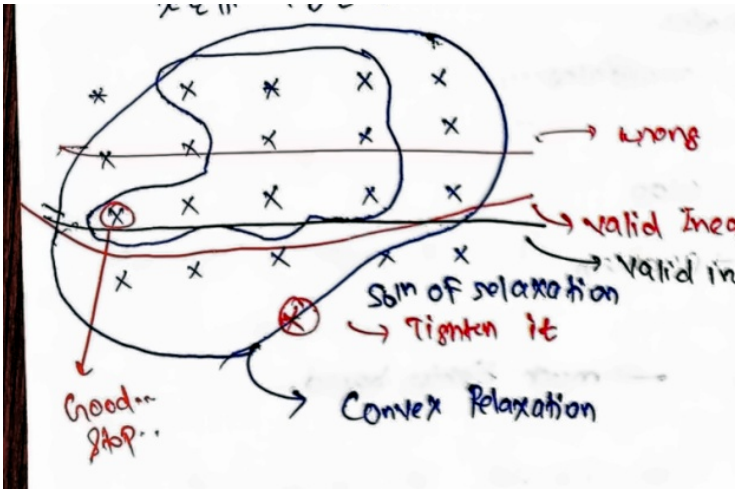


Figure 30: A large irregular region with scattered "x" points, a red line labeled "valid inequality" cutting a wrong region marked "wrong", another red line labeled "valid inequality" tightening near a circled optimal point, labeled "Good stop" and "Convex Relaxation" and "sum of relaxation → tighten it".

*** A linear inequality $d^T x + h^T y \leq k$ is a **cutting plane** for S, if it is valid for S and cuts off some given point, not in S.

How to find good cuts: lots of analysis. We have to find good cut.

Integer Linear Program:

General purpose cuts $\left\{$ Gomory mixed integer cut, Split cut, Lift and project cuts, Reduced split cut $\right\}$

Problem specific: Knapsack Cuts, Clique cuts, Flow cover, comb inequalities for TSP

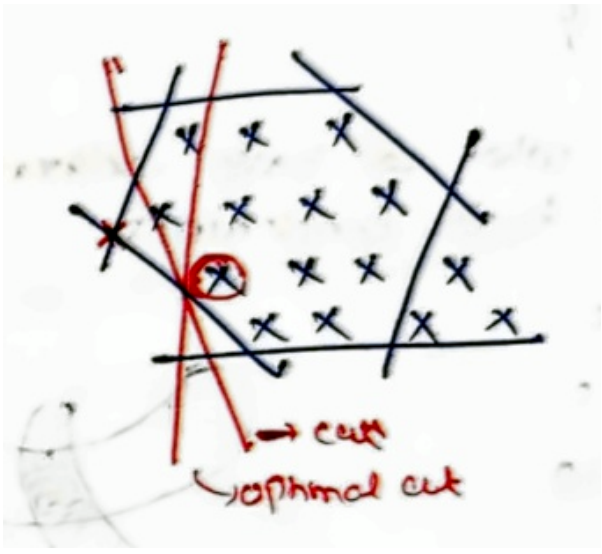


Figure 31: A polygon region with scattered "x" points, several cutting lines drawn through it converging near a circled point labeled "optimal cut" with an arrow "cut".

valid inequality → not disturbing feasible points

Given a model,

$$P\begin{cases}\text{Min } f(x,y)\\ g_i(x,y)\le 0,\ i=1,2\dots m\\ x\in\mathbb{R}^n,\ y\in\mathbb{Z}^n\end{cases}$$

- The set $S=\{(x,y):x\in\mathbb{R}^n, y\in\mathbb{Z}^p, g_i(x,y)\le 0, i=1,2\dots m\}$ is called feasible region of P
- An inequality $h(x,y)\le 0$ is said to be valid for S if all points of S satisfies $h(x,y)\le 0$ ***

Cuts From the Boolean Quadric Polytope:

$$\min c^Ty+y^TQy \quad \text{st } y\in\{0,1\}^n$$
$$=c_1y_1+\dots c_ny_n+q_{11}y_1^2+(q_{12}+q_{21})y_1y_2+\dots \quad \text{replace with } w_{12}$$

$$\min c^Ty+q_{11}y_1+q_{22}y_2+\dots+q_{nn}y_n$$
$$+q_{12}\underbrace{y_1y_2}_{w_{12}}+q_{21}\underbrace{y_1y_2}_{w_{12}}+q_{13}\underbrace{y_1y_3}_{w_{13}}+q_{14}\underbrace{y_1y_4}_{w_{14}}+\dots$$

st $w_{12}=y_1y_2$, $w_{13}=y_1y_3$, $w_{i,j}\in\{0,1\}$, $y_i\in\{0,1\}$

[circled note: $w_{12}\ge 0$; $w_{12}\le 1$ (not req.); $w_{12}\le y_1$; $w_{12}\le y_2$; $w_{12}\ge y_1+y_2-1$ ← corners are $[0,0,0]$, $[1,1,1]$, $[1,0,0]$, $[0,1,1]$, 3 active. When $n=2$: y_1 and y_2 only.]

$w_{13}\le y_1$, $w_{13}\le y_3$, $w_{13}\ge y_1+y_3-1$ — maybe we set some half values like $\frac{1}{2}w_{12}$ or something.

@ $n=3$: $w_{13}\ge 0$, $w_{13}\le y_1$, $w_{13}\le y_3$, $w_{13}\ge y_1+y_3-1$; $w_{23}\ge 0$, $w_{23}\le y_2$, $w_{23}\le y_3$, $w_{23}\ge y_2+y_3-1$ (complex...)

Table: $y_1,y_2,y_3,w_{12},w_{13},w_{23}$: rows (0,0,0,0,0,0), (0,0,1,0,0,0), (0,1,0,0,0,0), (0,1,1,0,?,1)...

Manfred Padberg: Triangle inequalities

$$*\ y_1+y_2+y_3-w_{12}-w_{13}-w_{23}\le 1$$

 It does not cut off integer feasible points. @ always valid → check logically

 It cuts off some non-integer points from e.g. $y_1=y_2=y_3=1/2$ ← allowed... $w_{12}=w_{13}=w_{23}=0\le y_2+w_{13}$

$$*\ w_{12}+w_{13}\le y_1+w_{23}$$
$$y_1=y_2=y_3=1/2\le y_3+w_{12}$$
$$w_{13}=w_{12}=1/2,\ w_{23}=0$$

@ don't add an inequalities blindly — solve and use some needed... # don't blindly add

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SDP Relaxation:

$$\min q_{11}x_1^2 + q_{12}x_1x_2 + q_{21}x_2x_1 + q_{13}x_1x_3 + q_{14}x_1x_4 + q_{22}x_2^2 + \dots q_{nn}x_n^2 \quad (n^2 \text{ terms})$$

(P) st (Index 1) $a_{11}x_1^2 + a_{12}x_1x_2 + a_{13}x_1x_3 + \dots a_{nn}x_n^2 = b_1$ (Index 2) $a_{11}x_1^2 + a_{12}x_1x_2 + \dots a_{nn}x_n^2 = b_2 \vdots$ (Index m) $a_{11}x_1^2 + a_{12}x_1x_2 + \dots a_{nn}x_n^2 = b_m$

$$l \le x \le u$$

Make it linear by adding some variables:

$$\min q_{11}x_{11} + q_{12}x_{12} + \dots q_{22}x_{22} + \dots + q_{nn}x_{nn}$$

st $a_{11}^1x_{11} + a_{12}^1x_{12} + \dots a_{nn}^1x_{nn} = b_1$ $a_{11}^2x_{11} + a_{12}^2x_{12} + \dots a_{nn}^2x_{nn} = b_2 \vdots a_{11}^mx_{11} + a_{12}^mx_{12} + \dots a_{nn}^mx_{nn} = b_m$

Non-linear part (I): $x_{11} = x_1x_1$, $x_{12} = x_1x_2$, $\vdots$, $x_{nn} = x_nx_n$

$$l \le x \le u$$

Sharp's Result: $X = xx^T$ is same as $X \ge 0$ PSD (call eigen values of $X \ge 0$), rank(X)=1

$(m+1) \times n^2 \rightarrow$ inputs → expressions

(I): $X = x \cdot x^T$

$$\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & & & \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} [x_1\ x_2\ \dots\ x_n] = \begin{bmatrix} x_1^2 & x_1x_2 & \dots \\ x_2x_1 & x_2^2 & \dots \\ x_3x_1 & x_3x_2 & \dots \\ \vdots & & \\ x_nx_1 & x_nx_2 & \dots \end{bmatrix}$$ each column is multiple of colu

(P) is same as ...

$$\min q_{11}x_{11} + q_{12}x_{12} \dots q_{nn}x_{nn}$$

$$\left.\text{st } a_{11}^1x_{11} + a_{12}^1x_{12} \dots a_{nn}^1x_{nn} = b_1\right\} \text{ Linear constraints (through } b_m)$$

$X = xx^T$] Quadratic constraints... $\approx X \ge 0 \rightarrow$ convex; rank(X)=1 → drop this → make relaxation... → Kla Semi-Definite Programming model, matrix is unknown, and it should be PSD

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \ge 0 \rightarrow \text{for this principal leading minors, det} \ge 0$$

LP vs **SDP**

LP: min c^Tx, $Ax = b$, $x \ge 0$ — solve by simplex

SDP: min $\langle c, x\rangle = c_{11}x_{11} + c_{12}x_{12} + \dots c_{nn}x_{nn}$, $\langle A, X\rangle = b$, $X \ge 0 \rightarrow$ solve by Interior Point method "Sam Burer" → cone, so convex

(Very diff...)

- Set S is a cone if for every $x \in S$, $\lambda x \in S \; \forall \lambda \geq 0$

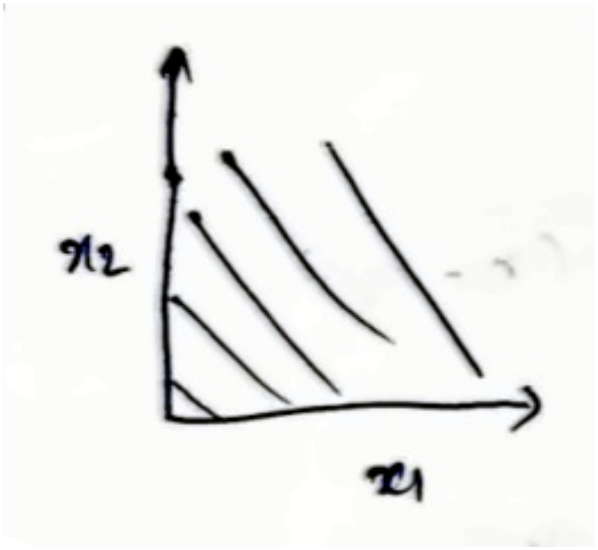


Figure 32: A shaded wedge region on (x_1, x_2) axes representing the LP feasible cone.

It is tight/perfect for all QCQPs with q^{n^3} Variables.

SDP relaxation + secant inequalities + bilinear relaxation ← these includes x; For a much better lower bound... For settings x and X:

$$\begin{bmatrix} 1 & x^T \\ x & X_{n \times n} \end{bmatrix} \geq 0$$

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$$x \in \mathbb{R}^{n \times n} \text{ - variable}$$
$$\min \langle c, X \rangle = \sum \sum c_{ij} x_{ij}$$
$$\text{st } \langle A_i, X \rangle \leq b_i, \; i = 1, 2 \ldots m$$
$$\boxed{X \geq 0} \leftarrow \text{convex constraint (+ve SD)}$$

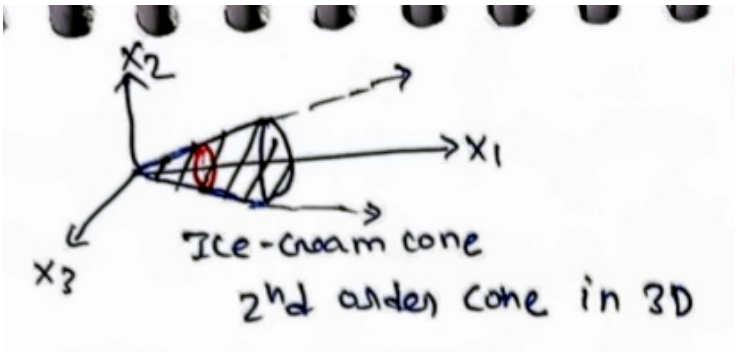


Figure 33: 3D axes x_1, x_2, x_3 with a shaded wedge region labeled “Ice-cream cone, 2nd order cone in 3D”.

equation: $\sqrt{x_2^2 + x_3^2} \leq x_1,\ x_1 \geq 0$

$$S = \{x \in \mathbb{R}^3 : x_1 \geq \sqrt{x_2^2 + x_3^2},\ x_1 \geq 0\}$$

If $\hat{x} \in S$, then $\lambda\hat{x} \in S\ \forall\lambda \geq 0$

QCQP (Quadratic Constrained Quadratic Progs)

$$\min x^T Q_0 x + a_0^T x$$

(P) st $\boxed{x^T Q_1 x + a_1^T x \leq b_1}$

$$x^T Q_2 x + a_2^T x \leq b_2$$
$$\vdots$$
$$x^T Q_m x + a_m^T x \leq b_m$$
$$l \leq x \leq u$$

 * If $Q_0, Q_1 \ldots Q_m$ are PSD then (P) is convex * If not → do bnb, relaxation, cutting etc…

 The constraint [box] *

*** $\sqrt{x_2^2 + x_3^2 + \cdots + x_n^2} \leq x_1,\ x_1 \geq 0$ $\Big\}$ n-dim cone

$$* \sqrt{(x_2 - 1)^2 + (3x_3 - 6)^2 + x_4^2 + \ldots x_n^2} \leq x_1$$

$$x_1 x_2 \geq 1 \rightarrow \text{not convex}$$
$$x_1 x_2 \geq 0 \rightarrow \text{now convex}$$

→ use this

$$4x_1 x_2 \geq 4$$
$$x_1 x_2 \geq 0$$

$$(x_1 + x_2)^2 - (x_1 - x_2)^2 \geq 4 \Leftrightarrow (x_1 + x_2) \geq \sqrt{2^2 + (x_1 - x_2)^2}$$

(Transformed cone)

 If Q is not PSD? → SDP slow, so less solvers: SeDuMi, MOSEK, CvxOpt

- If Q_1 has 1 -ve eigenvalue and some non-vity property, then it can be modeled as SOCP

Ex: $-x_1^2 + x_2^2 + x_3^2 \leq 0$, $x_1 \geq 0$ → 2nd order cone with one -ve eigen value

$$x_2^2 + x_3^2 \leq x_1^2 \rightarrow \text{non convex}$$

[Diagram: A double-cone (bowtie/hourglass) shape shaded, showing the non-convex region.]

$$x_2^2 + x_3^2 - x_1^2 \leq 0$$
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$x_2^2 + x_3^2 - x_1^2 \leq 0,\ x_1 \geq 0 \rightarrow \text{convex whether it looks non-convex}$$

$$4x_1x_2 \geq 4$$
$$x_1 + x_2 \geq 0$$

$\hookrightarrow (x_1 + x_2)^2 \geq 4 + (x_1 - x_2)^2$, also SOCP

Fujie & Kojima:

 Let (P) be a QCQP Let (P) have an SOCP convex relaxation of the form

$$\min x^T B_0 x + b_0^T x$$

$x \in \mathbb{R}^n$ st $x^T B_1 x + b_1^T x \leq \beta_1$ (Convex QCQP) $x^T B_2 x + b_2^T x \leq \beta_2 \vdots x^T B_k x + b_k^T x \leq \beta_k$

Then the SDP

(SDP) $\min\langle B_0, X\rangle + b_0^T x$ st $\langle B_1, X\rangle + b_1^T x \leq \beta_1 \vdots \langle B_k, X\rangle + b_k^T x \leq \beta_k$, $l \leq x \leq u$

has same optimal value as convex QCQP.

$$X - xx^T \geq 0 \Leftrightarrow \begin{bmatrix} 1 & x^T \\ x & X \end{bmatrix} \geq 0 \quad \text{same}$$

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Quadratic functions are 60% of solvers

Quadratic fn: $\min x^T Q_0 x + a_0^T x$ st $x^T Q_1 x + a_1^T x \leq b_1 \vdots x^T Q_m x + a_m^T x \leq b_m$ $l \leq x \leq u$

We can make it non-linear:

$$\min\langle Q_0, X\rangle + a_0^T x$$

st $\langle Q_1, X\rangle + a_1^T x \leq b_1$ $\langle Q_m, X\rangle + a_m^T x \leq b_m$ $l \leq x \leq u$

$$X = xx^T$$

$n \times n$ rank of $X = 1$ $X \geq 0 \rightarrow$ convex

$$\begin{bmatrix} 1 & x^T \\ x & X \end{bmatrix} \geq 0 \leftarrow \text{convex}$$

Now model becomes linear, SD problem. SDP gives a lower bound to original problem.

- SDP takes time to solve → slow, so we try some other ideas:

Idea 1) $\min x^T Q_0 x + a_0^T x$ st $Ax \leq b\ l \leq x \leq u$

rewrite $\rightarrow \min x^T(Q_0 + KI)x + a_0^T x - K \cdot x^T I x$

If we pick K big enough than most -ve eigen value of Q_0, the problem will be more convex.

Let σ_{min} = minimum eigen value of Q_0 (or) most -ve eigen value.

Then $K \geq |\sigma_{min}|$

$$\min x^T(Q_0 + KI)x + a_0^T x - \sum_{i=1}^{n} K x_i^2 \rightarrow \text{we don't need bilinear constraints}$$

$$Ax \leq b$$

$$l \leq x \leq b$$

→ Not always faster than original problem

DC Formulations → difference of convex formulations

If Q is big and non-convex → we can't solve

RLT (bounds play big role)

General QCQP: $\min x^T Q_0 x + a_0^T x$

$$x^T Q_1 x + a_1^T x \leq b_1$$

$$\vdots$$

$$x^T Q_m x + a_m^T x \leq b_m$$

$$l \leq x \leq u$$

We use: $X_{n\times n} = xx^T$, $x_{11} = x_1^2$, $x_{12} = x_1 x_2$, ⋮, $x_{ij} = x_i x_j$ → relax b/b drawing boxes

This becomes $\min\langle Q_0, X\rangle + a_0^T x$ → next page

Bilinear Part: $x_{12} = x_1 x_2$ Bound factors → $\{l_1 \leq x_1 \leq u_1,\ l_2 \leq x_2 \leq u_2$

$$(x_1 - l_1) \geq 0$$

$$(x_2 - l_2) \geq 0$$

multiple: $(x_1 - l_1)(x_2 - l_2) \geq 0$

$$x_1 x_2 \colon x_{12} + l_1 l_2 - x_1 l_2 - x_2 l_1 \geq 0 \rightarrow \text{linear constraint}$$

Using $(u_1 - x_1) \geq 0$, $(u_2 - x_2) \geq 0$: $x_{12} + u_1 u_2 - x_1 u_2 - x_2 u_1 \geq 0$ → called McCormick inequalities

Relaxation of linear:

try: $x_{11} = x_1^2$

$$(x_1 - l_1)(x_1 - l_1) \geq 0$$
$$x_{11} - 2l_1 x_1 + l_1^2 \geq 0$$
$$x_{11} \geq -l_1^2 + 2l_1 x_1$$

we will set second for $(x_1 - l_1)(u_1 - x_1) \geq 0$

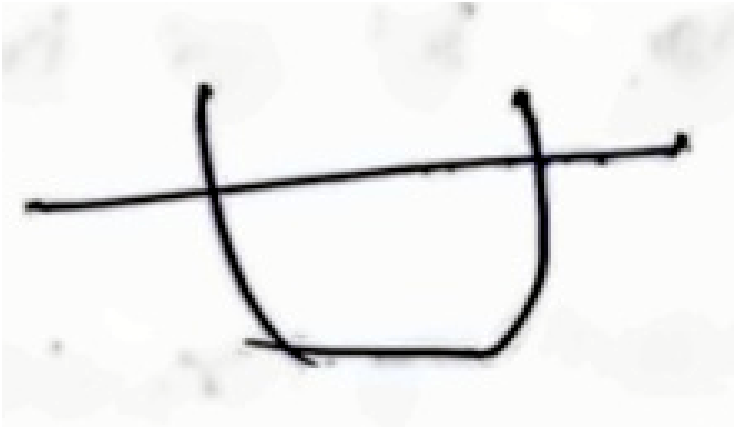


Figure 34: A parabola $x_{11} = x_1^2$ shape.

If we have some linear constraints

$$\min x^T Q_0 x + a_0^T x$$

$$l \leq x \leq u \rightarrow n \text{ dimension}$$
$$\{Gx \leq b\} \rightarrow k \text{ dimension}$$

$$(h_1 - g_1^T x) \geq 0$$
$$(x_1 - l_1) \geq 0$$
$$-h_1 l_1 - x_1 g_1^T x + h_1 x_1 + l_1 g_1^T x \geq 0$$

→ quadratic so rewrite as

$$x_1 \sum g_{1j} x_j$$
$$= \sum g_{1j} x_{1j} \qquad x_{12} \geq (u_1 u_2 - x_1 u_2 - x_2 u_1)$$

We can do lot of these,

$$3x_{12} - x_{23} + p \cdot x \leq s_1$$
$$3(x_1 u_2 + x_2 u_1 - u_1 u_2) - (\cdots) + p \cdot x \leq s_1$$

We can multiply again with other constraints and do one more round of RLT

$$(x_{12} + u_1u_2 - x_1u_2 - x_2u_1 \geq 0)$$
$$x_3 - l_3 \geq 0$$
$$x_{12} \cdot x_3 = x_{1,2,3}$$

More RLT more cutting. But we don't know will it converge. But we know it will make tighter.

- It was used initially for binary variables

$$\min c^T x$$
$$Ax \leq b$$
$$x \in \{0,1\}^n$$

$$x_1 + 5x_2 + x_3 \leq 1$$
$$x_1 \geq 0$$

$\rightarrow x_1(x_1 + 5x_2 + x_3 - 1) \leq 0$

$$x_{11} + 5x_{12} + x_{13} - x_1 \leq 0$$
$$x_{12} \leq x_1$$
$$x_{12} \leq x_2$$
$$x_{12} \geq x_1 + x_2 - 1$$

- multiply constraints with each other
- new model has lot of constraints
- No need of branch and bound, but in binary case we can get convex hull.

It does not have any square terms

$$x_1y_1 \geq 1$$
$$x_1 \geq 0$$
$$y_1 \geq 0 \text{ integer}$$

To find outer approximation convex hull of points, it becomes flatter and flatter as goes to ∞ segments

$$y_1 \geq 1$$
$$\vdots$$

Infinitely many linear constraints

$$S = \begin{cases} x_1y_1 + x_2y_2 + x_3y_3 + \dots x_ny_n \geq \gamma \\ x \geq 0 \\ y \geq 0 \text{ continuous} \end{cases}$$

convex hull of S?

Consider $\gamma = 6$, set S is Non-convex. take $x_1 = 3, y_1 = 2$ and other values $= 0$ $x_2 = x_3 = \dots x_n = 0$, $y_2 = y_3 = \dots y_n = 0$

we can show convex hull of S is given by ... divide both sides by γ and take $\surd$

$$\text{Conv}(S) = \left\{ \sqrt{\frac{x_1y_1}{\gamma}} + \sqrt{\frac{x_2y_2}{\gamma}} + \dots + \sqrt{\frac{x_ny_n}{\gamma}} \geq 1 \right\}$$

The result won't work when we make change in equation like x_2y_1 as x_1^2

- Every point feasible in top original formulation is also point in relaxation which is in bottom

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$$x_1y_1 \geq 1$$
$$x_1 \geq 0$$
$$y_1 \geq 0,\ y \to \text{continuous}$$

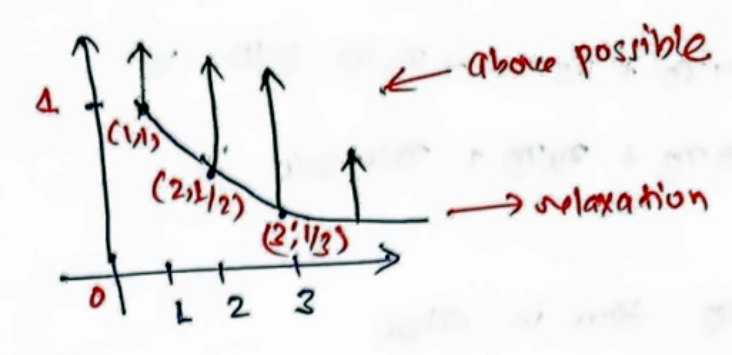


Figure 35: A hyperbola-like decreasing curve with sample points $(1, 1)$, $(2, 1/2)$, $(2, 1/3)$, arrows pointing up from each labeled "above possible" → relaxation.

Pure bilinear term y_1y_2, x_1y_2 Not allowed.

Due to no upper bound → we can't try... we can still use SDP.

A constraint of the form x^TQy is called a **pure bilinear term form**. There are no terms x_ix_j, y_ky_ℓ.

Example: (1) $x_1x_2 + x_3x_4 + x_5x_6 + x_7x_8 \geq 10$ □ (2) $x_1x_2 + x_3x_4 + x_2x_3 \geq 10$ □ (3) $x_1x_2 + x_2x_3 + x_1x_3 \geq 10$ □

every term in edge, can we divide the graph?

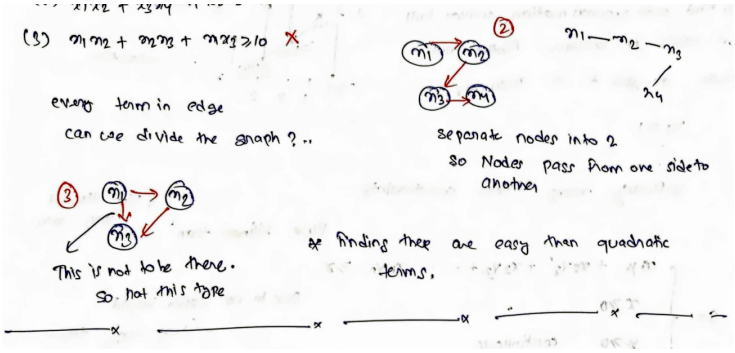


Figure 36: A 4-node graph ( ) with edges $m_1 \to m_2$, $m_3 \to m_4$, $m_2 \to m_3$ (or similar). Create nodes for variables, for terms; separate nodes into 2 so nodes pass from one side to another.

 $m_1 - m_2 - m_3$, x_4 branch — finding these are easy than quadratic terms.

 Triangle graph $m_1 \to m_2$, $m_2 \to m_3$, $m_3 \to m_1$ — this is not to be there, so not this type.

Let $S = \{(x, y) : x^T Q y + a^T x + b^T y + c \leq 0,\ 0 \leq x \leq 1,\ 0 \leq y \leq 1\}$

Result: Conv(S) can be written as a **second order cone**

closed form of SOC? → explore in paper

Let $S = \{(x, y) : x^T Q_1 y + a_1^T x + b_1^T y + c_1 \leq 0, \vdots, x^T Q_k y + a_k^T x + b_k^T y + c_k \leq 0, 0 \leq x \leq 1, 0 \leq y \leq 1\}$

Convex hull → Conv(S) is SOC representable.

We introduce this and use graphs to analyse their properties.

Given relaxation, how bad is relaxation?

if we take conic relaxation of S, how bad is that set.

$$x^T Q y + a^T x + b^T y + c \leq 0$$
$$0 \leq x \leq 1$$
$$0 \leq y \leq 1$$

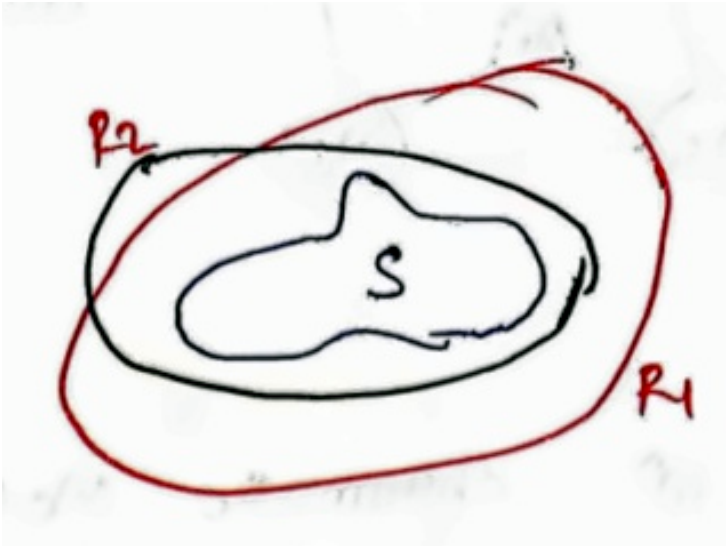


Figure 37: An irregular blob labeled S enclosed by a tighter red outline R_1 and a looser black outline R_2.

How to Quantify

 Find maximum of a point in R — for this we need to know S

 Take minimum function over $R, \in S$ - try different linear function and differentiate b/w optimizing over R_1 and S - find optimal value of R and S - if very diff answers then they are far off - if small result, then they are near.

 We look at function rather than set

Non-convex function of (x): find under-estimator of $f(x)$ over ... of $f(x)$ → need not tightly

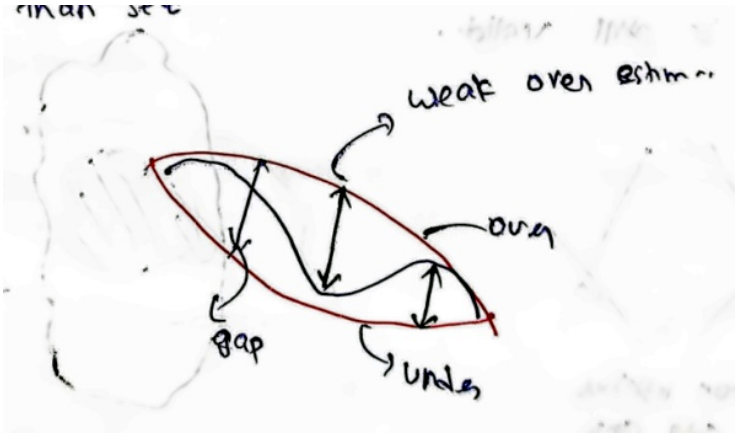


Figure 38: An irregular blob with a red oval circumscribing zigzag curves labeled “gap”, “over”, “under”, “weak over estim...”

→ measure gap. max gap → it is weak

analyse gap of simple mccormick eqn → gap is constant for this

Depending on shape of Q determines weakness (or) gap of McCormick relaxation.

x^TQx: [Diagram: A graph with 5 nodes m_4, m_2, m_1, m_5, m_3 connected by edges forming a shape with a triangle in the middle, "Chromatic No. = 3"]

Result says:

The weakness of McCormick Relaxation is related to the chromatic number of the associated graph.

- when No.=2, the McCormick relaxation is tight
- Chromatic No. → min. no. of colours such that some colour is not seen both sides.
- large chromatic No. $m = n$ then it will be very weak, gap is very high.

This is for single constraint only

For multiple quadratic constraints, we relax separately (or) we can add the constraint and relax it — it is still valid.

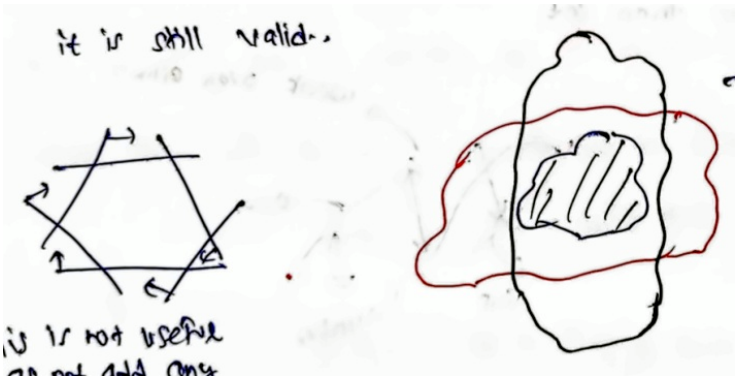


Figure 39: A hexagram-like star of crossing lines, "This is not useful as not add any value."

[Diagram: An irregular capsule-shaped region with a red outline and a small shaded shape inside labeled "quadratic".]

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Project Work: 15 minute presentation, your own work - diagram - example - what work, what doesn't? - assumptions → why? why not? - Can result [be] extended or simplified to other cases.. - Don't repeat the result..

Next 2-3 class → cutting plane → Linear, NL, Quad…

Cutting planes for MINLO: - QG algorithm - Padberg's triangle inequalities for boolean quadratic models.

MINLO: - Non-convex non-linearities - Convex non-linearities - Integer constraints

Integer Linear Case:

$$S = \left\{ y = \begin{matrix} y_1 \leq 2.8 \\ y_1 \in \mathbb{Z} \end{matrix} \right\}$$

Relax: $R = \{y : y_1 \leq 2.8, y_1 \in \mathbb{R}\}$

Cutting plane: $y_1 \leq 2$

$$y_1 \leq b,\ y_1 \in \mathbb{Z} \Rightarrow y_1 \leq \lfloor b \rfloor,\ y_1 \in \mathbb{Z}$$

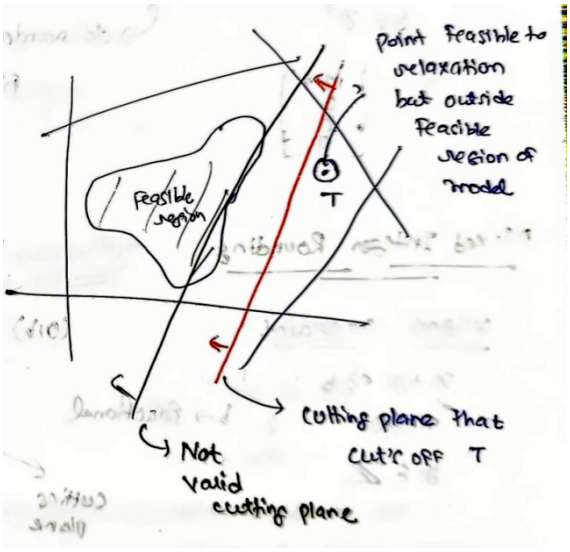


Figure 40: An irregular polygon with scattered “x” points, several crossing cutting lines, labeled “cutting plane not possible..”.

Ex: $6y_1 + 9y_2 \leq 14,\ y_1, y_2 \in \mathbb{Z}$

→ divide by 3: $2y_1 + 3y_2 \leq 5.6667$ → Now scale: $2y_1 + 3y_2 \leq 5$ → $\boxed{6y_1 + 9y_2 \leq 15}$ ← Round on method..

Chvátal and Gomory

$\square\ y_1 + 3.2y_2 - 2.3y_3 + 6y_4 = 12.4$

$$y \geq 0$$
$$y \in \mathbb{Z}^4$$

$$\hat{y} = \begin{bmatrix} 12.4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

→ do roundoff and apply above rule.. find cutting plane

Mixed Integer Rounding

original constraint: 1. $x_1 + y_1 \leq b$ 2. $x_1 \geq 0$ 3. $y_1 \in \mathbb{Z}$

$b \rightarrow$ fractional

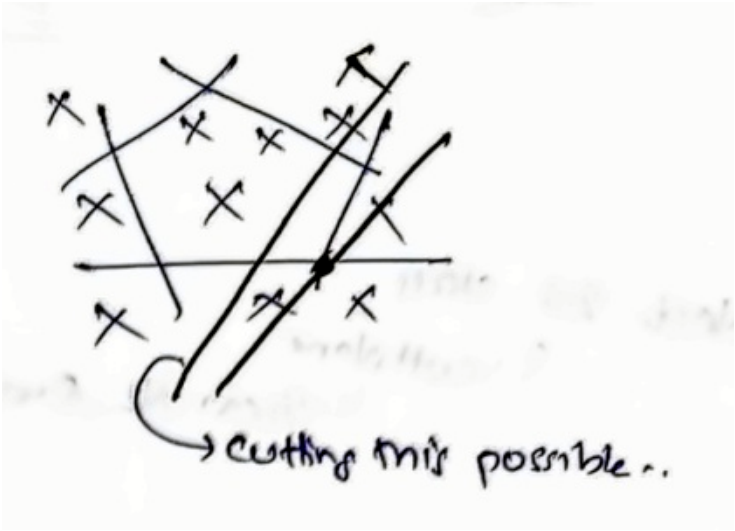


Figure 41: A plot of y_1 vs x_1, showing a family of parallel red lines at y_1-intercepts $(0, \lfloor b \rfloor + 1)$, $(0, b)$, $(b - \lfloor b \rfloor, \lfloor b \rfloor)$, and the constraint line $x_1 + y_1 \leq b$ crossing them, with a "cutting plane" labeled and "equation of cutting plane" pointing to it.

$$\frac{x_1}{f} + y_1 \geq \lceil b \rceil$$

Fractional part of $b = (b - \lfloor b \rfloor)$

After cutting + no need to cut...

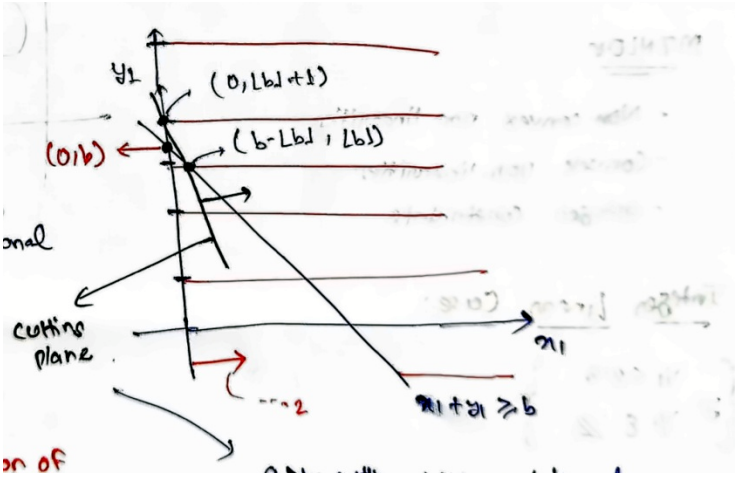


Figure 42: drawing from the page (no description in the notes)

$\lceil a_2 \rceil + f_2$, $\lceil a_3 \rceil + f_3$

$a_2 y_2 + a_3 y_3 \leq b + x_1$

$$y \geq 0$$

$$x \geq 0$$

$$y \in \mathbb{Z}^2$$

→ MIR (Mixed Integer Rounding) cut

$$\square \lceil a_2 \rceil y_2 + \left(\lceil a_3 \rceil + \frac{f_3 - f_0}{1 - f_0} \right) y_3 \leq \lfloor b \rfloor + \frac{x_1}{1 - f_0} \quad (b - \lfloor b \rfloor)$$

$$f_2 = a_2 - \lceil a_2 \rceil$$
$$f_3 = a_3 - \lceil a_3 \rceil$$

$$\square \left(\lceil a_2 \rceil + \frac{f_2 - f_0}{1 - f_0} \right) y_2 + \lceil a_3 \rceil y_3 \leq \lfloor b \rfloor + \frac{x_1}{1 - f_0}$$

also you can combine, and take maximum.

Generalization

$$y_1 + \sum_{i=2}^{p} a_i y_i + \sum_{i=1}^{n} a_i x_i = b$$
$$y \geq 0$$
$$y \in \mathbb{Z}^p$$
$$x \in \mathbb{R}^n$$

→ apply MIR cut and get Gomory mixed integer inequality.

Prop: (MIR, GG) - cheap cuts - Guaranteed to cut off fractional corner points - weak → convergence slow

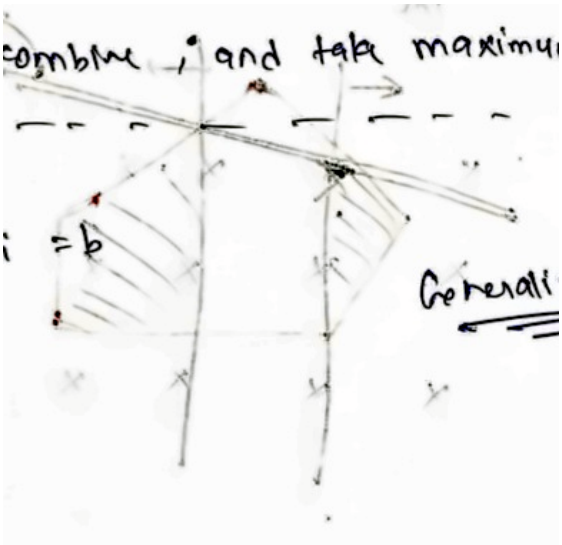


Figure 43: drawing from the page (no description in the notes)



Figure 44: A blob shape S with a point T inside it, labeled “Not possible” — you can’t separate a point from a set if the point is inside the set.

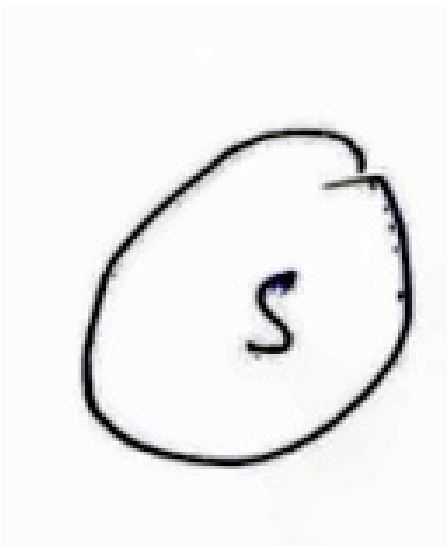


Figure 45: A blob S separate from a point T with a line between them, labeled “Separate S from T by a hyperplane.. or linear”.

Separation Problem

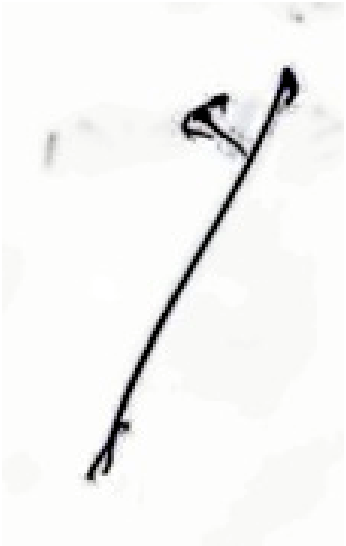


Figure 46: A plot with y_1 axis, several roughly-horizontal squiggly bands labeled S_1 (upper) and S_2 (lower), and a straight red line crossing through them, illustrating “relax into two parts”.

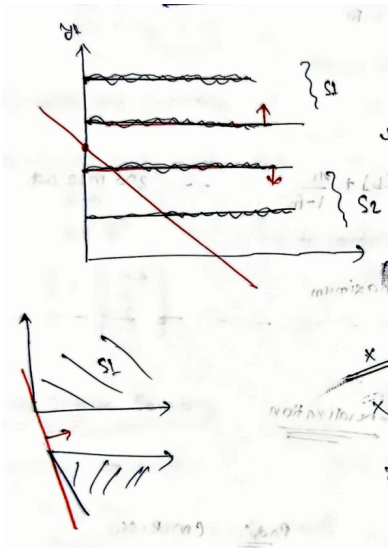


Figure 47: A cone of rays from the origin labeled S_1, with a red separating line.

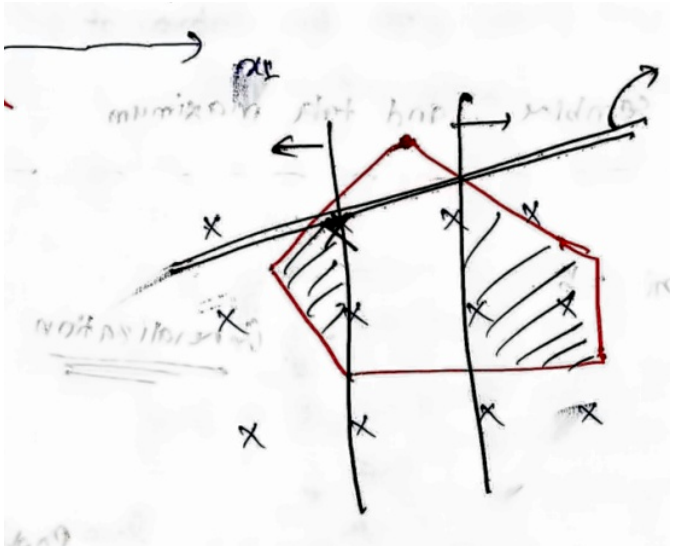


Figure 48: An irregular pentagon region outlined in red with scattered “x” points, split by two vertical lines into hatched sub-regions, with a diagonal line labeled “Split cuts…”.

02/04/26

Split inequalities:

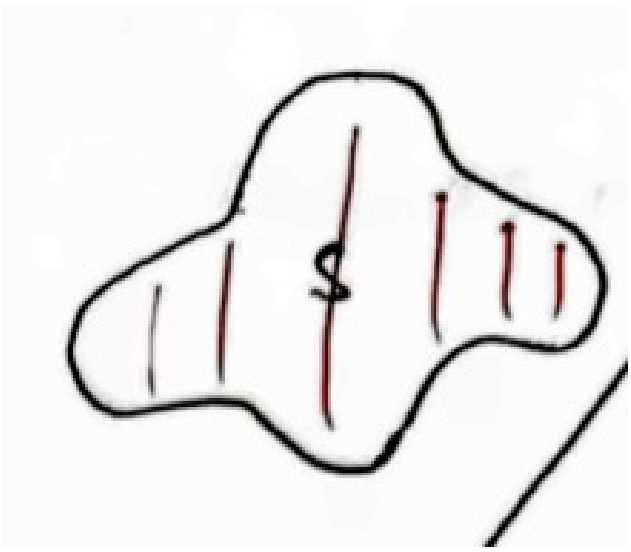


Figure 49: An irregular blob S with several vertical hatch lines inside, and an arrow pointing to a line labeled $\alpha x \leq \beta$, captioned “To check the inequality is valid for S”.

If $\beta^* \leq \beta \rightarrow$ Then it is valid for S ($\alpha x \leq \beta$)

Find, $\beta^* = \max \alpha x$ st $x \in S$

Let $S = S_1 \cup S_2$

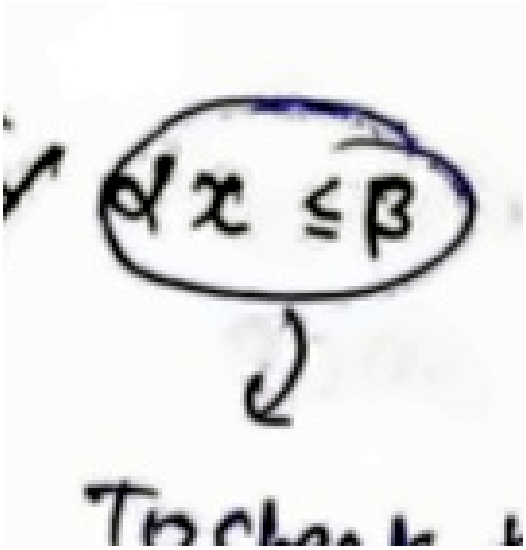


Figure 50: Two blob shapes S_1 and S_2 overlapping, joined by a line.

Suppose $\alpha x \leq \beta$ is valid for both S_1 and S_2.

Then $\alpha x \leq \beta$ is valid for S

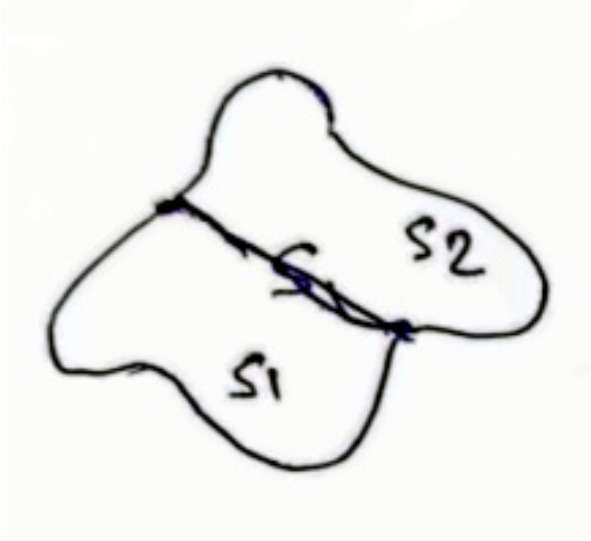


Figure 51: Two separate blobs S_1 and S_2 with a line $\alpha x \leq \beta$ between them.

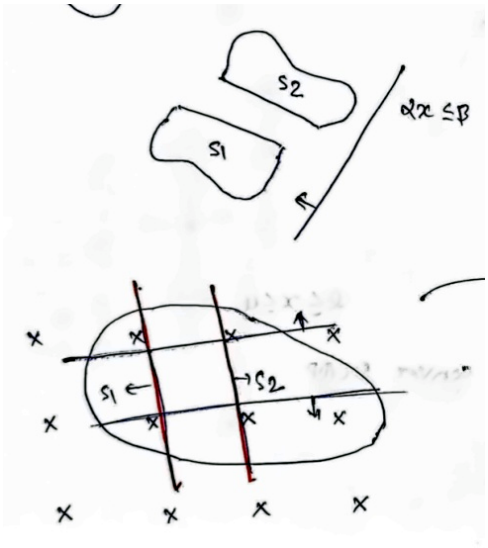


Figure 52: An irregular region with scattered "x" points, divided by two roughly parallel lines into S_1 and S_2 sub-regions.

→ how to find cutting planes? → Split → in different ways

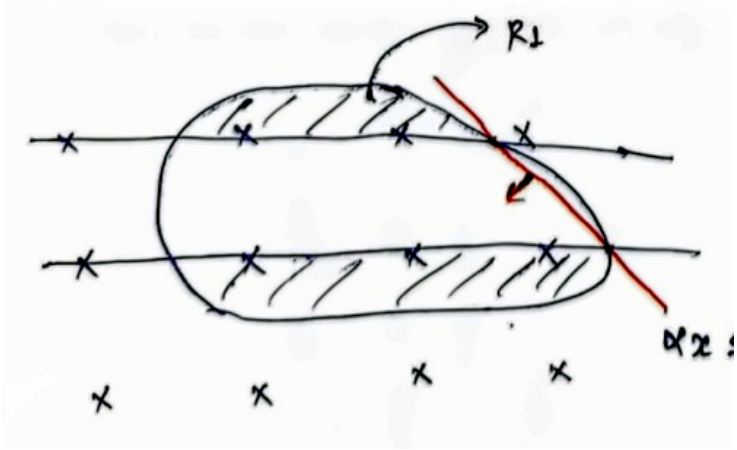


Figure 53: An irregular hatched region with a red line labeled $\alpha x \leq \beta$ cutting through it, region R_1 marked, "find inequality valid for both half".

Mathematical

$$\min c^T x + d^T y$$
$$\text{st } g_i(x,y) \leq 0,\ i = 1, 2 \dots m$$
$$x \in \mathbb{R}^n$$
$$y \in \mathbb{Z}^p$$

$$S = \{(x,y) : x \in \mathbb{R}^n, y \in \mathbb{Z}^p,\ g_i(x,y) \leq 0,\ i = 1, 2 \dots m\}$$

$$R = \{(x, y) : x \in \mathbb{R}^n, y \in \mathbb{R}^p,\ g_i(x, y) \leq 0,\ i = 1, 2 \dots m\}$$

Find (α, β) st it is valid for S but not R

Step □ * Pick a disjunction (OR) eg. $\boxed{y_1 \leq 0 \text{ or } y_1 \geq 1}$ → pick this eg. $y_5 \leq 4$ or $y_5 \geq 5$ eg. $y_5 + y_6 + y_7 \leq 1$ or $y_5 + y_6 + y_7 \geq 2$

□

$$S_1 = \{(x, y) : x \in \mathbb{R}^n, y \in \mathbb{Z}^p,\ g_i(x, y) \leq 0,\ i = 1, 2 \dots m,\ y_1 \leq 1\}$$
$$S_2 = \{\dots,\ y_1 \geq 2\}$$
$$S = S_1 \cup S_2$$

Step □ Relax → R_1 and R_2

$$R_1 = \{\dots, x \in \mathbb{R}^n, y \in \mathbb{R}^p, \dots, y_1 \leq 1\}$$
$$R_2 = \{\dots, y \in \mathbb{R}^p, \dots, y_1 \geq 2\}$$

Step □ Find $\alpha x + \gamma y \leq \beta$ valid for both R_1 and R_2

→ use R_1, R_2 ... to get to Red line. → There are lots of lines like the Red line (infinite)

Stubbs & Mehrotra, 1999

Give the convex hull of $R_1 \cup R_2$ with the assumptions.. (i) $y \to$ binary (ii) R's are bounded (iii) disjunctions used like $y_i = 0$ or $y_i = 1$

Convex hull is given by:

$$(x^1, y^1) \in R_1$$
$$(x^2, y^2) \in R_2$$
$$(x, y) = \lambda^1(x^1, y^1) + \lambda^2(x^2, y^2)$$

with $\lambda^1 + \lambda^2 = 1$, $\lambda^1 \geq 0$, $\lambda^2 \geq 0$

$$M = \{(x_1^1, y_1^1, x_2^2, y_2^2, \lambda_1, \lambda_2^2) :$$
$$g_i(x^1, y^1) \leq 0 \to \text{cx}$$
$$g_i(x^2, y^2) \leq 0 \to \text{cx}$$
$$x = \lambda^1 x^1 + \lambda^2 x^2 \to \text{NCX}$$
$$y = \lambda^1 y^1 + \lambda^2 y^2 \to \text{NCX} \to \text{very difficult part}$$
$$\lambda^1 + \lambda^2 = 1 \to \text{cx}$$
$$x^1, \lambda^2 \geq 0 \to \text{cx}$$
$$\left.\begin{matrix} y_1^1 = 0 \end{matrix}\right\} \to \text{cx}$$
$$\left.\begin{matrix} y_1^2 = 1 \end{matrix}\right\}$$

Assuming $y_i = 0$ or $y_i = 1$. Form new constraint in place of difficult one:

$$a = \bar{x}_1 + \bar{x}_2, \quad b = \bar{y}_1 + \bar{y}_2 \leftarrow \text{use some trick}$$

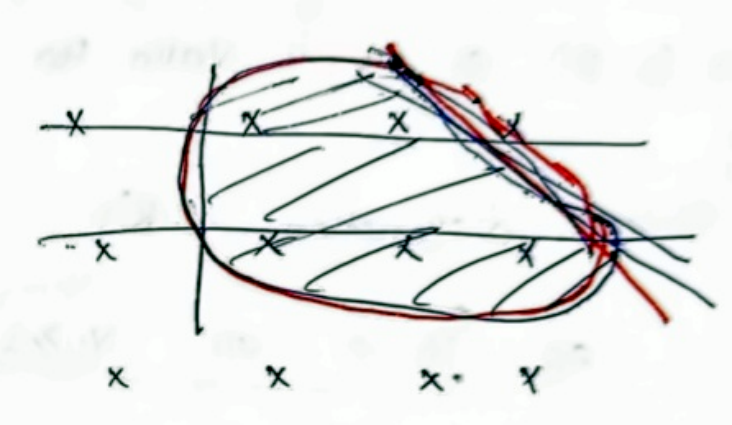


Figure 54: drawing from the page (no description in the notes)

Projection of M on (x, y) space is convex hull of $R_1 \cup R_2$

$$M_{\text{convex}} = \{\bar{x}^1, \bar{y}^1, \bar{x}^2, \bar{y}^2, x, y, \lambda^1, \lambda^2 :$$
$$\lambda^1 g_i \left(\frac{\bar{x}^1}{\lambda^1}, \frac{\bar{y}^1}{\lambda^1}\right) \leq 0$$
$$\lambda^2 g_i \left(\frac{\bar{x}^2}{\lambda^2}, \frac{\bar{y}^2}{\lambda^2}\right) \leq 0$$
$$x = \bar{x}^1 + \bar{x}^2$$
$$y = \bar{y}^1 + \bar{y}^2$$
$$\lambda^1 + \lambda^2 = 1,\ y_i \geq 0\Big\}$$

→ all are convex now..

Perspective function

$$p(\cdots) = \begin{cases} \lambda^1 g_i \left(\frac{\bar{x}^1}{\lambda^1}, \frac{\bar{y}^1}{\lambda^1}\right) & \lambda^1 > 0 \\ 0 & \lambda^1 = 0 \end{cases}$$

Upcoming: - Intersection cuts - Cuts for non-convex Quadratic - Heuristic

Next Thursday: 2-3 min per team, short presentation. Next to next → doubt class.

06/04/26

Topics to be discussed: - Intersection Cuts - Cuts for non-convex quadratics - Heuristics

Intersection Cuts:

$$\min c^T y$$
$$Ay \leq b$$
$$y \in \mathbb{Z}^p$$

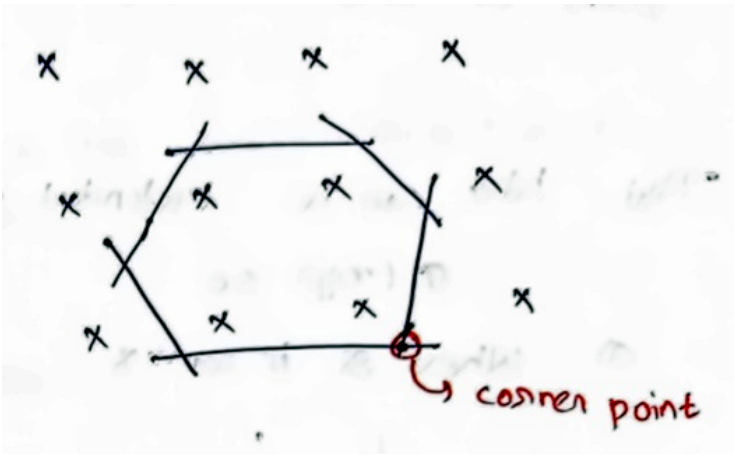


Figure 55: A hexagonal region with scattered "x" integer points, one vertex circled and labeled "corner point".

(1) Solve LP relaxation, where we will be at corner

(2) If we drop other constraints which are not active, we end up only 2 constraints. [Diagram: scattered x points with two crossing lines meeting at a point labeled "we still get same point".]

(3) If we look at integer points present, it is translation of cone. This is relaxation of original set.

- We can chop off some points by looking at cone.

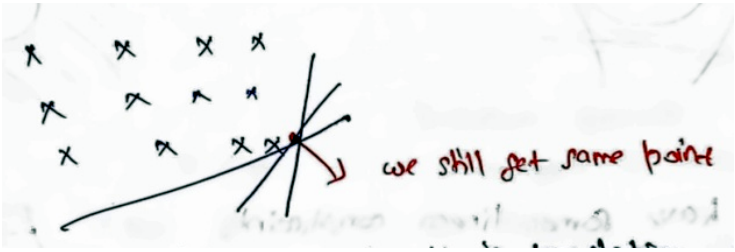


Figure 56: scattered x points with two crossing lines and a horizontal line through a circled point, labeled "valid cut for original problem too", "how to get these cuts".

we create a disjunction as all points on one of these sides

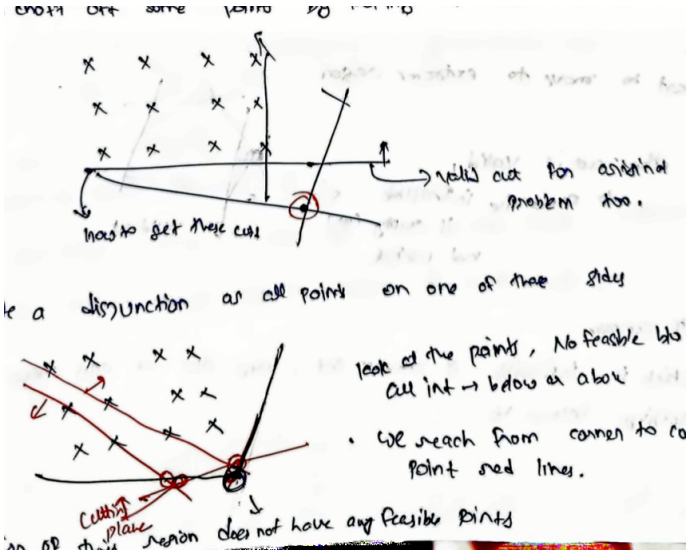


Figure 57: scattered x and + points with two red diagonal lines converging near a circled point.

look at the points, No feasible b/w hand lines, all int → below or above - we reach from corner to corner of point red lines.

Interior of this region does not have any feasible points

Cutting plane

- We can choose any corner point initially but the disjunction should be cutting the corner point (or) the corner point should be b/w the 2 disjunctions to cut-off.
- If the starting point is not a corner of feasible region but infeasible region.

This is better as all feasible points are within the cone.

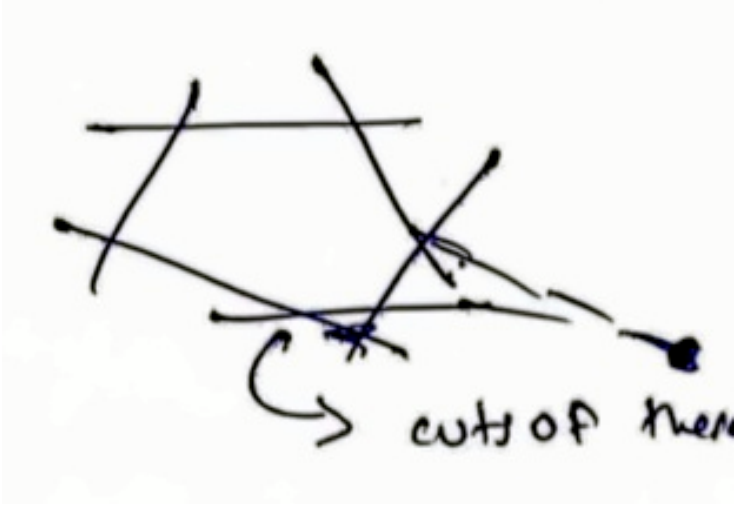


Figure 58: A crossing-lines sketch with a circled “basic soln point” and an arrow “cuts off here are much deeper”.

This idea can be extended to non-convex too. $g_i(x, y) \leq 0$

a) Where g_i is convex — we take relaxation of this case [Diagram: A hexagon with an inscribed circle.]

b) Suppose g_i is concave, $-m_1^2 - m_2^2 \leq -1$, outside circle [Diagram: A spiral/circle shape with hatching around it, "we can't chop off the point".]

- Suppose we have some linear constraints, since hard we ignore concave
- We may set corner point $\otimes$ if it is infeasible region then we are done
- if not like, we start to move to extreme region

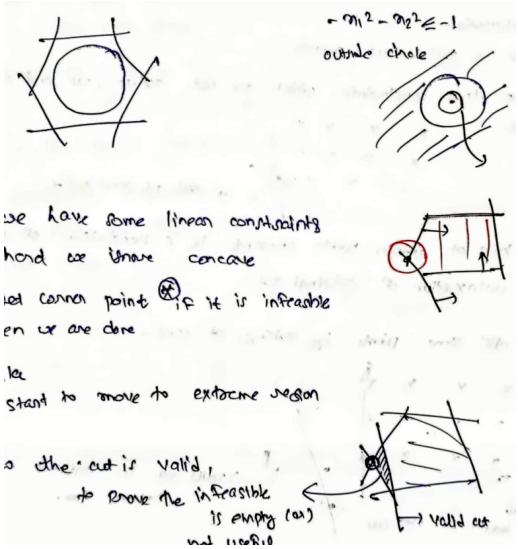


Figure 59: A quadrilateral with hatching and a circled point, arrow labeled "valid cut".

How to show the cut is valid, to prove the infeasible is empty (or) not useful.

Since $g(\cdot)$ is concave * The part which is infeasible is convex set, using this we can show there is nothing below it..

Only works for $g_i(\cdot)$ is concave

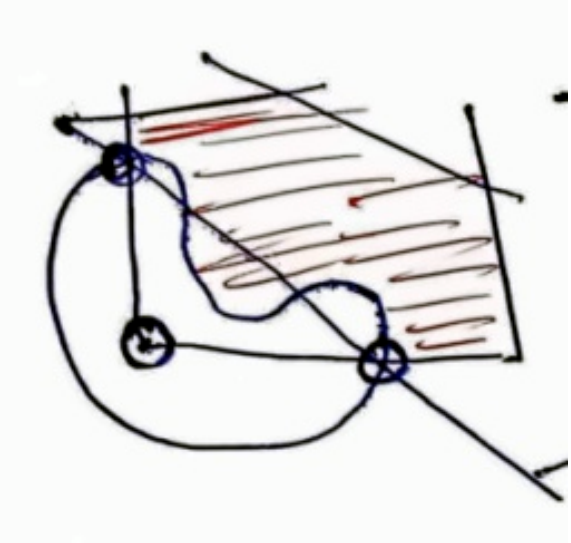


Figure 60: A hatched infeasible region with three circled points on its boundary, arrow: "travel 2 sides of corner point and reach till the edge of infeasible region then draw a cut."

→ Not a valid cut as it cuts feasible red [region]

- This works for high dim. 3D → 2 constraints → move 3 direction to find points → draw hyperplane that intersect sphere
- N dim → N direction → N points

Provided constraint is concave if works:

$$x^T Q' x \leq b_i$$

if Q is PD → it is convex if Q is ND → concave

If neither PD nor ND: indefinite symmetric matrix can be decomposed to difference of 2 terms, using eigen value decomp.

DCF: $x^T(Q'^+ - Q'^-)x \leq b_1$ (Q'^+: PD, Q'^-: ND) ← diff of convex fn

also as: $x^T Q'^+ x - w \leq b_1 \rightarrow$ convex concave: $w - x^T Q'^- x \leq 0$

Tangent → convex Intersection points → concave - In practical using simplex of intersection points → one is prone → many solvers do not use it unless $Q \rightarrow$ diagonal..

$$\min x^T Q x + c^T x$$
$$0 \leq x \leq 1 \longrightarrow \text{No integer}$$

→ introduce new variable

$$\min \sum Q_{ij} x_{ij} + \sum Q_{ii} y_i + \sum c_i x_i$$
$$x_{ij} = x_i x_j \ \forall i \neq j$$
$$y_i = x_i^2 \ \forall i$$
$$0 \leq x_i \leq 1$$

→ In case of only boolean $x \in \{0,1\}$

Padberg's triangle inequalities

$$\min \sum Q_{ij}x_{ij} + \sum c_i x_i$$

st $x_{ij} = x_i x_j \ \forall i \neq j$

$$x_i \in \{0,1\} \ \forall i$$

Convex hull of ⌘ is same as convex hull of $x_{ij} = x_i x_j$, $0 \leq x_i \leq 1$

Result by Burer and Letchford..

Finding good upper bounds: 07-04

$$\min f(x,y)$$
$$g_i(x,y) \leq 0, \ i = 1,2 \ldots m$$
$$x \in \mathbb{R}^n$$
$$y \in \mathbb{Z}^p$$

bcz goal is to find primal soln not dual

- Upper bound → find a feasible point as fast as possible, as low obj. value as possible } Primal heuristic

Good upper bound - prune the tree early - give a good solution in limited time

- Gurobi may have 60-70 heuristics..

Heuristics - Starting Heuristics (Don't need a feasible point to start) — try to construct a soln from scratch → kla construction heuristics - Improvement Heuristics (start with given a feasible soln → find better solns)

MINLOs: Today's class: - Heuristic for unconstrained models - for model with many feasible solutions - for models with few solutions (difficult task)

△ another way to classify

Heuristic → General purpose (structure independent) → Class specific (only Quadratic, Contin. Non Convex Quadratic.., Conv. MIQP, conic integer models) Imp° → Application specific (Portfolio opti., power flow problems, cutting stock, VRP, packing, Quad. assignment..)

△ $\min f(x,y)$, $x \in \mathbb{R}^n$, $y \in \mathbb{Z}^p$

(1) Algo 1 1. Pick a random point (x,y) with $x \in \mathbb{R}^n$ and $y \in \mathbb{Z}^p$ 2. Check if it is best 3. Go to 1 (until value not improves) → stopping criteria is important

works but not so good... → we have good idea @ after many trials

Neighbourhood search: → do

(2) Algo-2 1. Solve continuous relaxation 2. Round y to nearest integer → Good if convex problem relaxation is there

also couple with Neighbourhood search → Try different rounding

$$y = \begin{bmatrix} 0.6 \\ 0.4 \\ 0.2 \\ 0.5 \end{bmatrix} \rightarrow \text{we get..}$$

$\triangle$ min $f(x, y)$

$$0 \leq x \leq 1$$
$$y \in \{0, 1\}^p$$

$$\min f(x, y), \quad \sum_{i=1}^{p} y_i = k, \quad y_i \in \{0, 1\},\ i = 1 \ldots p$$

(1) Algo-1 → same mostly, most soln on x.. is 0 and 1 not b/w...

Continuous Non-convex Quadratic Models:

$$\min \frac{1}{2} x^T Q x + c^T x$$
$$\text{st } \frac{1}{2} x^T Q_i x + a_i^T x \leq b_i$$
$$l \leq x \leq u$$

⊙ **Multistart NLP Heuristic** - Give it to a local gradient based solver

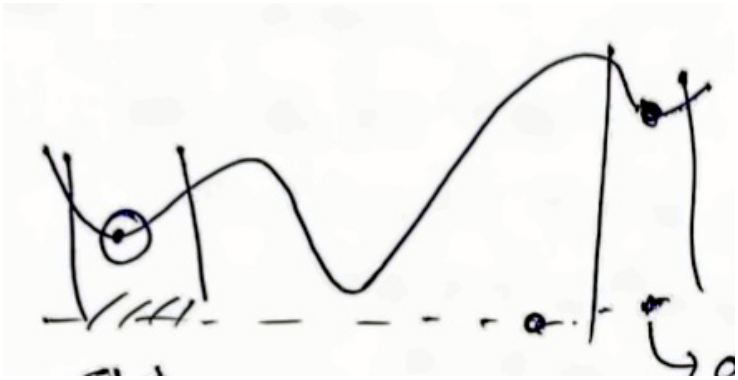


Figure 61: A wavy non-convex curve with a local minimum circled at “Start” and another point marked further right with an arrow.

→ also give them another point, if not good then previous take another point.

⊙ **Undercover heuristic** - You pick some x values, suppose $p = 100$ → pick 90 and fix for some values

$$x_1^2 + 5x_1x_2 + 7x_3x_4 + x_1x_3 - 2x_1x_4 \leq 1$$

1st iteration: $x_1 = 2$, $x_3 = 1$ → next change values and solve NLP; take x_1 as anything, x_4 (change variable)

- How to know which variable to pick and what values… → graph → trial and error

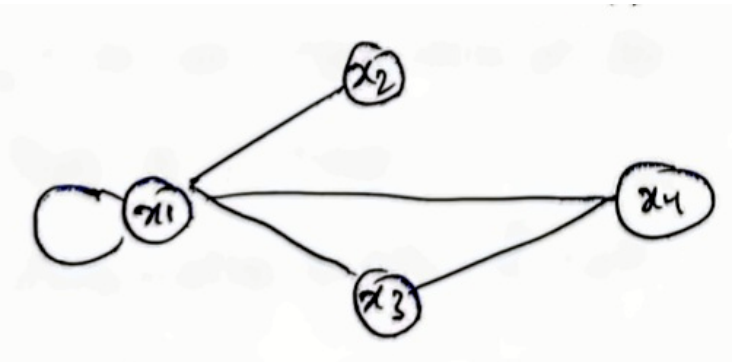


Figure 62: A graph with node x_1 (with a self-loop) connected to x_2, x_3, x_4.

… pick smallest node so that every set is covered.. - Find smallest node (edge) cover for this graph so that every edge is covered.. (x_1, x_3), (x_1, x_4) → solve..

Heuristics for MINLO 13/04/26

Initial / construction Heuristic - Feasibility Pump (expensive) → rounding (sophisticated) - RINS - Local Branching

MILP Case: (Pure binary)

$$\min c^T x$$
$$Ax \leq b$$
$$x \in \{0,1\}^n$$

 Solve the LP: $\min c^T x$, $Ax \leq b$, $0 \leq x \leq 1$ → $\bar{x}$

→ if fractional → round it to nearest integer → Not always work

Simple rounding of $\bar{x} = \begin{bmatrix} \lfloor x_1 \rfloor \text{ if } x_1 < 0.5, \text{o.w. } \lceil x_1 \rceil \\ \vdots \\ \lfloor x_i \rfloor \text{ or } \lceil x_i \rceil \end{bmatrix} \Leftrightarrow \min \|x - \bar{x}\|_1$, $x \in \{0,1\}^n$, $Ax \leq b$

→ this will always give integer but hard..

 rounding of $\bar{x}$: $\min \sum_{i=1}^{n} |x_i - \bar{x}|$, $Ax \leq b$, $0 \leq x \leq 1$ // $\bar{x} \rightarrow$ LP relaxation soln

eg. $\bar{x} = \begin{bmatrix} 0 \\ 1 \\ 0.4 \\ 0.6 \\ 0.8 \\ 0.1 \end{bmatrix}$, round($\bar{x}$)= $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ if works → good

$$\min |x_1 - 0| + |x_2 - 1| + |x_3 - 0| + |x_4 - 1| + |x_5 - 1| + |x_6 - 0|$$

st $Ax \leq b$, $0 \leq x \leq 1$

becomes $\min x_1 + (1 - x_2) + x_3 + (1 - x_4) + (1 - x_5) + x_6$ st $Ax \leq b$, $0 \leq x \leq 1$

if $\bar{x} = \begin{bmatrix} 0.1 \\ 0.9 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ → again round off gives same soln → we have to stop heuristic and flip some variables..

Convex MINLO Case (Pure Binary)

$$\min c^T x + d^T y$$
$$g_i(x, y) \leq 0$$
$$x \in \mathbb{R}^n$$
$$y \in \{0, 1\}^p \rightarrow \text{continuous}$$

 Solve NLP relaxation → get $\bar{x}, \bar{y}$ → round off $\bar{y}$

⊛ You can also solve MILP relaxation

 $\min \sum_{i=1}^{p} |y_i - \text{round}(\bar{y})| \Leftrightarrow \min \sum |y_i - \text{round}(y_i)|$ st $g_i(x, y) \leq 0$, $0 \leq y \leq 1$ → st ∇ inequalities at $(\bar{x}, \bar{y})$, $0 \leq y \leq 1$

$\Leftrightarrow$ MILP $\{\vdots\}$, $y \in \{0, 1\}^p$

Non-convex MINLO Case (Pure Binary) → Leave it...

RINS Heuristic (Relaxed-Induced Nbhd Search) → Improvement

Convex MILO: Let $(\hat{x}, \hat{y})$ be known solution of MINLO

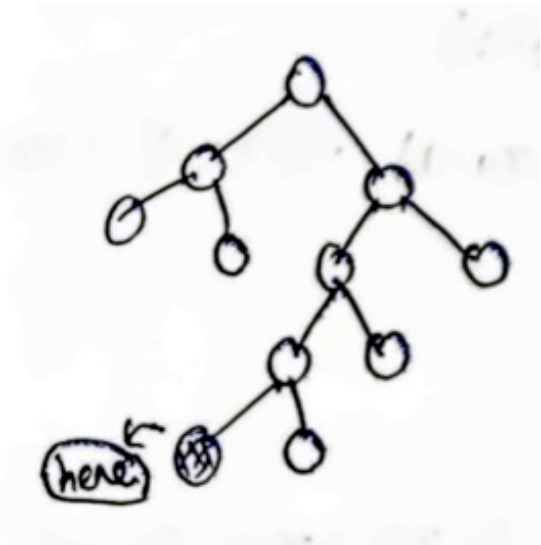


Figure 63: A branch-and-bound tree with several circle nodes, one node circled and labeled “here”.

 Solve NLP relaxation of that node → $(\bar{x}, \bar{y})$, $\bar{y} \notin \mathbb{Z}^p$ → search nbhd of it → if majority of equal → randomly fix, $(\hat{y}_i = \bar{y}_i)$

 comp. $\hat{y}$ and $\bar{y}$ → all those same → fix it

 Solve the smaller MINLO → don't solve fully (Node limit) put time limits ex. 5 sec.. and some other parameters..

Local Branching (Improvement)

Let $(\hat{x}, \hat{y})$ be a known solution, $y \in \{0,1\}^p$

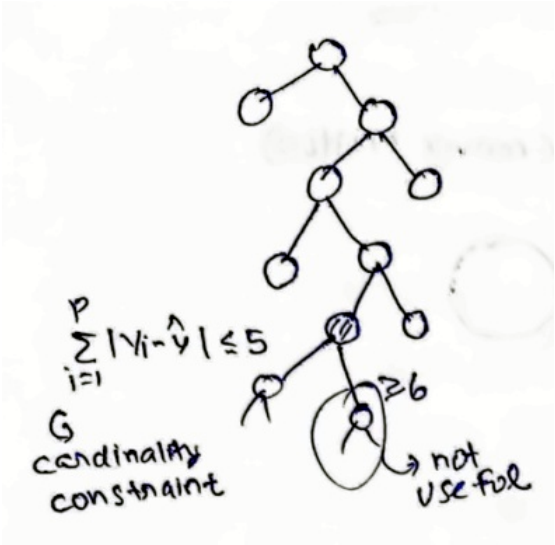


Figure 64: A branch-and-bound tree with several nodes, one node circled and connected to two children, one labeled "≥ 6" and marked "not useful".

$$\sum_{i=1}^{p} |y_i - \hat{y}_i| \leq 5$$

→ cardinality constraint

Local Jump: solving LPs takes too long **Idiot Crash:** Initial / construction

```
\min \bcancel{c^Tx}
```

Warning: equation could not be typeset; shown as written.

$$Ay \leq b$$
$$y \in \mathbb{Z}^p$$

$$\min \sum w_i (b_i - Ay)^+$$

$$3y_1 + y_2 - y_3 \leq 7 \rightarrow y_1 \text{ should be small}$$
$$-y_1 + 2y_2 + y_3 \geq 9 \rightarrow y_1 \text{ should be small}$$
$$y_2 + y_3 + y_4 = 5$$
$$y_3 - y_4 + y_6 \leq 8$$

$$0 \leq y \leq 10$$

set $y_1 = 0$

Recall: 16-04-26

- computation graphs

Convex MINLO - ECP - Quesada Grossmann - Duran and Grossmann - Non-linear branch and bound - Bonmin

Extended cutting plane (convex MINLO)

- Separability
  - Perspective reformulation
  - Automatic reformulation

 Bound tightening Heuristics → Local B0 → Random → RINS → Multi start → Feas Pump → Under Cover

 SDP relaxation Boolean Quadratic Polytopes

 Cuts → MIR, GG, Split cuts - Secants McCormick ineq. RLT ○ SOC? Contn dist..